

Population agglomeration, borrowed size and urban economic growth in China

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ABSTRACT

This study constructed a theoretical framework and measurement model for investigating the effect of population agglomeration on urban economic growth. This framework accounts for both intracity and intercity agglomeration economy and is based on a comprehensive set of indicators of population agglomeration and urban economy. Two hypotheses were tested using data from 285 prefecture-level cities in China from 2005 to 2020, with the data corroborated against data from over 500 Chinese provincial and municipal statistical yearbooks, 4560 statistical bulletins, natural elements such as relief degree of land surface, and DMSP/OLS and NPP/VIIRS matched nighttime lights imagery. We focused on the effects of intercity agglomeration economy, namely borrowed size and agglomeration shadow effects. Our results indicated that intracity population agglomeration had a positive effect on urban economic growth. This effect was moderated by spatial and policy-related factors. In addition, the borrowed size effect was stronger than the agglomeration shadow effect among Chinese cities. Intracity population agglomeration had a significant positive spatial spillover effect. The high degree of agglomeration of neighboring cities positively affected a city's economic growth. It was interesting to note that the impact coefficient of intracity agglomeration dropped by about 0.1 with the inclusion of intercity agglomeration elements.

1. Introduction

Borrowed size and agglomeration shadow are two hypotheses that explain the influence of intercity population agglomeration on economic growth. These hypotheses suggest that intercity population agglomeration may lead to two opposing economic phenomena. The first is a “borrowed size” phenomenon—where an area around an agglomeration possesses a large economy and high level of infrastructure that are disproportionate to the area's small size—and the second is an “agglomeration shadow” phenomenon—where resources and economic activity in a neighboring area are displaced out into the agglomeration (Burger et al., 2015). In other words, small cities adjacent to a large city gain access to the labor market, capital, and services in the large urban agglomeration (and thus gain borrowed size) while avoiding the problem of overcrowding (Alonso, 1973; Camagni et al., 2015). However, the development of small cities may also be hindered because they live under the agglomeration shadow of a large city (Burger et al., 2015;

Fujita et al., 1999). Most studies on this topic have empirically argued that the effect of borrowed size between cities is greater than that of the agglomeration shadow (Camagni et al., 2015; Kudelko & Musial-Malago, 2022; Liu & Chen, 2017; Sun & Ding, 2016). In China, some scholars have observed substantial agglomeration shadow effects in monocentric structure urban agglomerations, such as the Chengdu–Chongqing urban agglomeration and the Central Plains urban agglomeration, in addition to in the poverty belt around Beijing (Chen & Sun, 2017; Yang et al., 2022). In terms of policy formulation and urban development, the development model of China's national metropolitan areas, in which a core city drives the development of surrounding small and medium-sized cities, exemplifies the positive spillover effect on neighboring cities through borrowed size. By contrast, the development patterns of declining regions, such as provincial border zones and peripheral circles of urban agglomerations, reflect an agglomeration shadow. Therefore, exploring the general pattern of the influence of intercity population agglomeration on economic growth at the level of

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prefectural cities in China is crucial to assessing the effects of borrowed size and agglomeration shadow on Chinese cities.

Moreover, uncovering the general effects of population agglomeration on economic growth in Chinese cities remains challenging. Studies have demonstrated that urban economic growth is intrinsically related to agglomeration economy—a spatial pattern in which population, capital, and other factors concentrated distributed in a specific space, generating spillover effects such as sharing, matching, and learning (Chen, 2022; Duranton & Puga, 2004). In particular, by considering both macro (city-level) and micro (individual-level) units of analyses, scholars such as Edward Glaeser, Gilles Duranton, and Diego Puga have demonstrated the significant promoting effect of a population agglomeration on the economic growth of cities, particularly cities in Europe and the United States (Combes et al., 2008; Duranton & Puga, 2004; Glaeser & Gottlieb, 2009; Glaeser & Resseger, 2010). Some studies have demonstrated that, because of the high population density of Chinese cities (Ding, 2004), population agglomeration and urban economic growth in China have an inverted U-shaped relationship, suggesting that excessive population sizes may diminish urban economic growth in the country (Au & Henderson, 2006). Thus, multiple indicators and robust data must be used in analyses of whether and how population agglomeration influence urban economic growth, and densely populated cities must be included in the analysis. However, a comprehensive set of theoretical frameworks and empirical models considering intracity and intercity agglomeration is lacking.

Therefore, based on the “borrowed size” hypothesis, we constructed a theoretical framework and measurement model. This framework considered both intracity agglomeration and intercity agglomeration. Using a 2005–2020 multi-source corroborated dataset and a comprehensive set of indicators of 285 prefectural-level cities in China, we investigated the impact of urban population agglomeration on urban economic growth. This paper proposes two research questions: (1) does intracity population agglomeration significantly contribute to urban economic growth? and (2) are there spatial spillovers from this urban population agglomeration? If there are spatial spillovers, is the impact of intercity population agglomeration stronger in terms of “borrowed size” or “agglomeration shadow”?

Our work is innovative in three aspects: first, we constructed a theoretical framework and measurement model that considered both intracity agglomeration and intercity agglomeration, which clarified the relationship between intracity agglomeration, borrowed size, and agglomeration shadow. We validated the rationality of the theoretical model by taking 285 prefectural-level cities in China as examples. Previous related studies have either considered single intracity agglomeration or intercity borrowed size or agglomeration shadow. Our study combining intracity and intercity agglomeration economies is an advance on previous single studies. Second, we have endeavored to obtain a robust set of mutually corroborating datasets from multiple data sources. These data sources include more than 500 Chinese provincial and municipal statistical yearbooks, 4560 statistical bulletins, data from the China Intellectual Property Office and the Geospatial Data Cloud such as relief degree of land surface, slope, and river network density, and DMSP/OLS and NPP/VIIRS matched nighttime lights imagery data. This work is a breakthrough in terms of data quantity and difficulty of data acquisition compared with previous studies. It also ensures the rationality and accuracy of our empirical evidence. Third, we select a comprehensive set of indicators to measure population agglomeration and economic growth. These indicators are population density, initial city size, whether it belongs to an urban agglomeration (a dummy variable indicating whether a city belongs to Beijing–Tianjin–Hebei, Yangtze River Delta, or Pearl River Delta urban agglomerations), gross domestic product (GDP) per capita, and per capita urban land revenue—the latter two of which are indicators of economic level. This work is another advance over the single indicator system.

The remainder of this paper is arranged as follows. Section 2 reviews

the literature concerning agglomeration economy and urban economic growth. Section 3 proposes our research framework and hypotheses that considers both intracity agglomeration and intercity agglomeration, the theoretical formula, and the data sources. Section 4 presents the empirical results on the effect of population agglomeration on urban economic growth. Section 5 offers a discussion and conclusion.

2. Literature review

2.1. Intracity agglomeration and economic growth

Various indicators of urban agglomeration have been adopted in the literature, and industrial specialization (Combes, 2000; Duranton & Puga, 2000; Henderson, 1997) and diversity (Glaeser et al., 1992; Henderson, 2003; Jacobs, 1969) are two traditionally used indicators. Henderson (1997) demonstrated that specialization is more influential than diversity on industrial employment growth in the United States. Glaeser et al. (1992) described that industrial diversity is a key driver of urban economic growth in 170 US cities. Combes (2000) revealed that diversity also contributed to employment growth in the service sector compared to specialization in France. In more recent studies, the spatial Gini coefficient (He et al., 2007), Thiel index (Brühlhart & Sbergami, 2009), and location quotient (Yu & Liu, 2021) have been used to measure urban industrial agglomeration. However, in modern cities, where science and technology and information technology are developing rapidly and service industries are coming in large numbers, how to measure the level of urban overall agglomeration has begun to become the focus of academic attention (Ciccone & Hall, 1996; Glaeser, 2011). In measuring the level of urban overall agglomeration, the academic community initially used the population size indicator (Au & Henderson, 2006; Kanemoto et al., 1996). However, there is a clear endogenous problem between population size and economic growth, and for this reason academics have focused on the important role of population density (Ciccone & Hall, 1996). Furthermore, Ciccone (2002) argued that when agglomeration externalities are related to physical distance, agglomeration externalities depends on the density rather than scale of agglomeration; thus, compared with population size, population density is more reflective of the agglomeration economic level. Moreover, increasing urbanization has made intercity connections increasingly important (Meijers et al., 2016), and urban agglomeration, as an advanced form of spatial organization in the mature stage of urban development (Zhou et al., 2015), has become an important carrier of regional economic development. Thus, whether a city belongs to an urban agglomeration also can indicate its level of urban overall agglomeration (Cheng et al., 2020).

Studies using population size as an indicator have demonstrated that urban agglomeration drives urban economic growth. Glaeser and Resseger (2010) reported a 13 % increase in labor output per capita for every twofold increase in population size in US cities. Duranton (2016) reported an elasticity coefficient of 5.1 % in the positive relationship between population size and urban wage growth in Colombia during 2008–2012. Abhishek et al. (2017) analyzed census data and showed that initial population size significantly influenced urban economic growth in India. Xi et al. (2019) demonstrated that wages are associated with a premium in big cities in China; specifically, according to 2012–2014 data from a survey of the labor force, every twofold increase in city size is associated with a 2.3 % increase in wages. Studies using population density as an indicator have demonstrated a positive relationship between agglomeration and economic growth in cities (Duranton & Puga, 2020; Song et al., 2020). Ciccone and Hall (1996) demonstrated that every twofold increase in employment density was associated with a 6 % increase in productivity in United States during 1988–1989. Ciccone (2002) also analyzed 1986 and 1987 data from five European countries and reported that every twofold increase in employment density was associated with a 4.5 %–5 % increase in labor productivity. Faberman and Matthew (2016) reported that every

twofold increase in urban density was associated with a 3 % increase in average wages among workers in US firms between 1992 and 1997. [Chen, Li, et al. \(2018\)](#) demonstrated that population density influenced the economic growth (indicated by per capita urban land revenue) of 35 provincial capitals and planned cities in China from 2005 to 2013, where every 10 % increase in population density was associated with a 6.2 % increase in per capita urban land revenue. [Song et al. \(2020\)](#) also found that for every 1 % increase in urban population agglomeration, the GDP per capita would increase by 1.743 % in the Beijing–Tianjin–Hebei urban agglomeration during 2005–2016.

However, some studies have discovered that population size and population density exhibit an inverted U-shaped relationship with urban economic growth ([Au & Henderson, 2006](#)). For example, [Kanemoto et al. \(1996\)](#) discovered a Goldilocks population range of 200,000–400,000 people in relation to productivity growth in Japan; within this range, a twofold increase in city size was associated with a 25 % increase in productivity. This inverted U-shaped relationship has also been reported in many studies on China. For example, [Au and Henderson \(2006\)](#) found that 51 %–63 % of cities were smaller than the optimal size in relation to per capita labor output during 1990–1997 in China. Furthermore, by analyzing 2003–2008 data for 286 prefecture-level cities in China, [Ke and Zhao \(2014\)](#) reported that the optimal city size was 6.29 million people in relation to output per capita, and that 19 % of cities were larger than this size in 2008. [Su et al. \(2017\)](#) analyzed 2006–2010 data and demonstrated that the optimal population density of Chinese cities was 13,000 persons/km² in relation to output value per capita, and that 6.7 % of cities had a higher-than-optimal population density in 2011. In addition, using data from 108 prefecture-level cities in China, [Cheng et al. \(2020\)](#) demonstrated that being located in an urban agglomeration contributes to economic growth (correlation coefficient = 2.97 %).

Location ([Chen, Li, et al., 2018](#)), government policies ([Abhishek et al., 2017](#); [Brühlhart & Sbergami, 2009](#)), and natural resources ([Glaeser & Gottlieb, 2009](#)) affect the relationship between agglomeration and economic growth. For example, [Abhishek et al. \(2017\)](#) reported that the relationship between agglomeration and growth is stronger in capital cities than in noncapital cities in Indian. [Brühlhart and Sbergami \(2009\)](#) demonstrated that economic openness positively moderates the agglomeration–growth relationship in an analysis of 1960–1996 data on 105 countries. [Glaeser and Gottlieb \(2009\)](#) found that the average January temperature can effect population agglomeration and thus effect economic growth in US cities. [Chen, Li, et al. \(2018\)](#) demonstrated that the positive influence of population density on economic growth was the strongest to weakest in the eastern, central, and western regions of China, respectively.

2.2. Intercity agglomeration and economic growth

Regarding the spillover effects and externalities of agglomeration (i.e., intercity agglomeration), [Ciccone \(2002\)](#) distinguished between internal agglomeration and external agglomeration as early as exploring the impact of population density on economic growth. Borrowed size and agglomeration shadow effects are two examples of intercity agglomeration economy and have been observed in previous studies ([Burger et al., 2015](#); [Camagni et al., 2015](#); [Phelps et al., 2001](#)).

The empirical study of intercity agglomeration effect mainly presented two opposing conclusions. First, there existed a borrowed size effect ([Phelps et al., 2001](#)). For example, [Camagni et al. \(2015\)](#) observed that second-tier cities enjoyed borrowed size effects through intercity networks and thus outperformed the first-rank cities in terms of economic growth in Western Europe during 1995–2006. [Sun and Ding \(2016\)](#) also observed borrowed size effects when analyzing 2000–2010 data from 108 cities of different tiers in China's Yangtze River Delta region. Specifically, on average, a small city had 0.74 % higher economic growth for every 100 km it was closer to a major city. [Kudelko and Musial-Malago \(2022\)](#) found that municipalities within functional

areas undergo greater economic development when they are situated close to large cities (especially core megacities) with a higher level of economic development in Poland. In an analysis of 2016–2018 data on 75 innovation pilot cities in China, [Hou \(2022\)](#) discovered that borrowed size effects can arise through high-speed-rail linkages between small and large cities; these borrowed size effects are in the form of spillovers of knowledge, human capital, and investment from large to small cities. This effect was strongest in second-tier cities with 1–3 million people.

Second, there existed an agglomeration shadow effect ([Fujita et al., 1999](#)). For example, [Burger et al. \(2015\)](#) analyzed 2006–2007 data on 2794 cities in North-West Europe and discovered that larger cities grew economically at the expense of other cities in the region. In other words, smaller cities surrounding larger cities were more likely to face the agglomeration shadow. [Chen and Sun \(2017\)](#) observed a similar phenomenon in the Beijing–Tianjin–Hebei urban agglomeration from 2000 to 2010, which gave rise to a “poverty belt around Beijing and Tianjin.” Similarly, [Zhang and Chen \(2022\)](#) reported that counties connected by high-speed rail to and situated within 97–195 km from a large city were under an agglomeration shadow; on average, GDP per capita decreased by 2.6 percentage points after a county gained high-speed rail access to other (larger) cities. However, [Cuberes et al. \(2021\)](#) suggest that the positive (borrowed size) versus negative (agglomeration shadow) effects of agglomeration on economic growth may vary by space and time. Specifically, those authors analyzed 1840–2017 US data at county and metropolitan levels and demonstrated that population growth in small cities was hindered by proximity to large urban centers in 1840–1920 but promoted by such proximity after 1920. [Yao et al. \(2020\)](#) found that small and medium-sized cities were more likely to borrow size within urban agglomeration, while large cities would generate agglomeration shadow by adopting the data of 283 prefectural-level cities of China in 2003–2016.

In summary, specialization and diversity are the classical indicators of agglomeration economic. With the increasing sophistication of technology and the growth of the service sector, population size, and population density, which measure the overall economic level of urban agglomeration have begun to attract the attention of urban researchers. The existing body of scholarly evidence indicates the positive influence of agglomeration economy on urban economic growth ([Melo et al., 2009](#)). Nonetheless, agglomeration economy may diminish the growth of cities to the right of the peak of the inverted U-shaped curve ([Au & Henderson, 2006](#)), especially in densely populated cities in China. Considering the aforementioned findings, this study sought to clarify whether population agglomeration drives or hinders growth by proposing a novel theoretical framework accounting for both intracity agglomeration and intercity agglomeration (i.e., borrowed size and agglomeration shadow effects) and by solving two methodological problems in the literature. The first problem is the endogeneity problem between population agglomeration and economic growth (measured as GDP per capita) ([Su et al., 2017](#)). This study solved this problem by using both per capita urban land revenue and GDP per capita to measure economic growth and adopting the generalized method of moments (GMM) with an added lagged term for income per capita to validate the robustness of the regression result ([Brühlhart & Sbergami, 2009](#)). We further used natural elements such as relief degree of land surface, slope, and river network density as instrumental variables for population agglomeration ([Glaeser & Gottlieb, 2009](#)), which tested by two-stage least squares. The second problem is the lack of verification for data from Chinese statistical yearbooks in previous studies on China, especially those demonstrating an inverted U-shaped relationship. This study solved this problem by corroborating Chinese statistical yearbook data against nighttime lighting data, which indicate that population density is arguably more accurate than official statistics ([Ch et al., 2020](#); [Chen, Li, et al., 2018](#)). More importantly, the borrowed size effect and agglomeration shadow effect provided a theoretical foundation for investigating intercity agglomeration economies. However, there is no

consensus on the impact of intercity agglomeration on urban economic growth. This study focused on the debate on whether agglomeration shadow or borrowed size effects are more dominant in China (with its dense and urbanized population), which added diversity to an otherwise Eurocentric literature (Burger et al., 2015; Camagni et al., 2015).

3. Theoretical framework and data sources

3.1. Theoretical framework and hypotheses

Studies have conventionally focused on the economic effects of intracity agglomeration rather than intercity agglomeration (Ciccone & Hall, 1996; Duranton, 2016; Glaeser & Gottlieb, 2009). However, agglomeration economies also generate some spillover effects on neighboring cities (Alonso, 1973; Ciccone, 2002; Wang et al., 2022). Thus, based on the traditional intracity agglomeration economy studies, we constructed a framework that considered both the impacts of intracity agglomeration and intercity agglomeration economic (specifically, borrowed size and agglomeration shadow effects) on urban economic growth. We propose two research hypotheses as follows:

Hypothesis 1. For cities, population agglomeration can substantially promote economic growth, although congestion may inhibit growth.

Regarding the impact of population agglomeration on urban economic growth, most studies have documented that intracity population agglomeration has a significantly positive impact on urban economic growth (Glaeser & Resseger, 2010; Melo et al., 2009; Zhang & Liu, 2008). Although some scholars argued that intracity agglomeration also suffers from problems such as congestion and environmental pollution, which inhibited economic growth, relevant empirical evidence has shown that intracity agglomeration economies boost more than hinder (Camagni et al., 2015; Ciccone, 2002; Duranton & Puga, 2020; Lu, 2017; Pan et al., 2024). Empirically, most studies have used population density and initial city size as a proxy for intracity agglomeration and discovered that population agglomeration boosts urban economic growth (Abhishek et al., 2017; Chen, Li, et al., 2018; Faberman & Matthew, 2016; Song et al., 2020). Of course, a few studies discovered an inverted U-shaped relationship between population agglomeration and economic growth (Su et al., 2017). However, these studies have doubtful results

due to endogeneity and unreasonable data acquisition.

For this purpose, we constructed the intracity agglomeration conceptual diagram (Fig. 1-a). Intracity population agglomeration generates positive agglomeration forces that promote urban growth (solid arrows) and negative crowding forces that hinder urban growth (dashed arrows). Positive agglomeration forces include elements such as population agglomeration, industrial agglomeration, and location advantages, whereas negative crowding forces include elements such as congestion, pollution, and crime. Crucially, the effects of positive internal agglomeration forces exceed those of negative crowding forces, indicating that intracity population agglomeration substantially contributes to urban economic growth. This net positive growth pattern is illustrated by the change in Fig. 1-a from a solid hexagon to a black dashed hexagon.

Hypothesis 2. Urban population agglomeration has a positive spatial spillover effect, and the high degree of agglomeration of neighboring cities positively affects a city's economic growth. That is, the borrowed size effect is stronger than the agglomeration shadow effect.

Alonso (1973) first conceptualized the borrowed size effect, specifically, the small cities adjacent to a large city can access the benefits from the labor market, capital, and services in the large urban agglomeration while avoiding the problem of overcrowding. Meijers and Burger (2017) reported that the borrowed size effect can also apply to large cities near an even larger one. Moreover, Wang et al. (2015) discovered a mutually beneficial development pattern of neighboring cities borrowing from each other, resulting in positive spillover effects in cities adjacent to those with higher economic levels and negative spillover effects in cities adjacent to those with lower economic levels. However, Fujita et al. (1999) argued that large cities could inhibit the growth of neighboring small cities, as large cities tend to attract labor, capital, and other resources from neighboring small cities during intercity interactions, called agglomeration shadow.

Some empirical studies have demonstrated that the borrowed size effect is stronger than the agglomeration shadow between cities (Kudelko & Musial-Malago, 2022; Phelps et al., 2001; Sun & Ding, 2016; Yao & Wu, 2020). For example, Kudelko and Musial-Malago (2022) discovered that the closer city to the core mega-cities within functional areas had better economic development in Poland during 2003–2018.

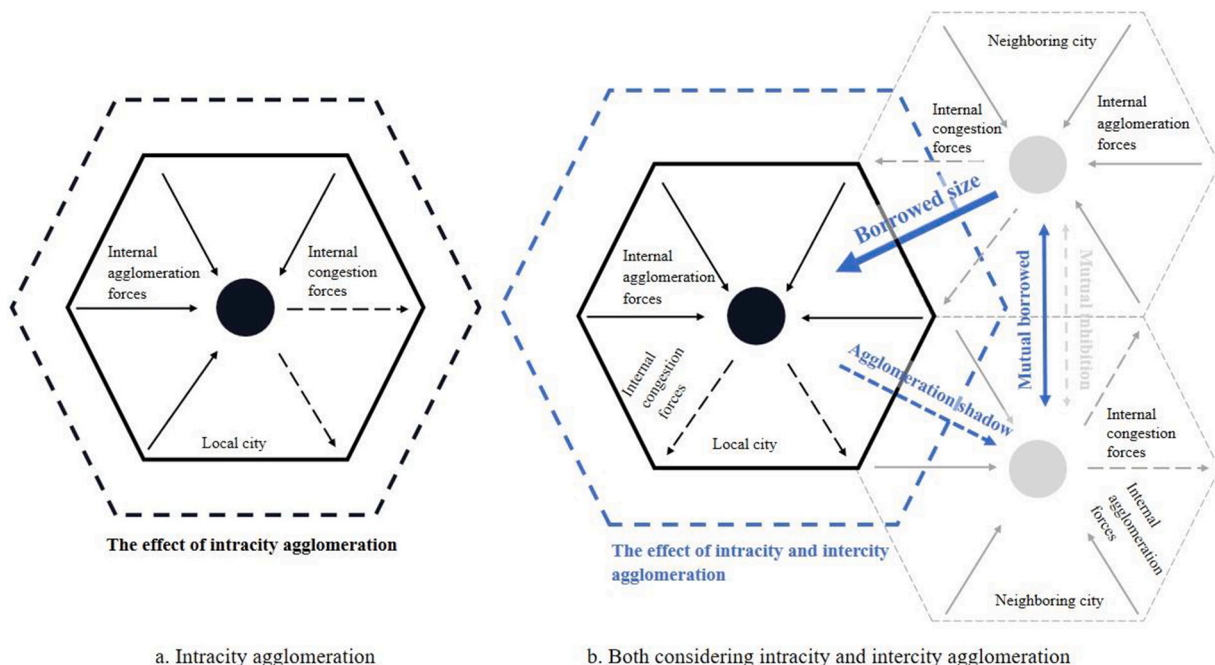


Fig. 1. Evolution of empirical ideas concerning the effect of urban agglomeration on urban economic growth.

Additionally, in examining the influence of the geographic boundaries of agglomeration economies on the urban economic growth of Chinese cities, Liu and Chen (2017) demonstrated that small cities can enhance their productivity by borrowing the size and functions of neighboring large cities, suggesting that the borrowed size effect is greater than the agglomeration shadow effect. Despite the greater strength of the borrowed size effect, the agglomeration shadow effect is not negligible (Burger et al., 2015; Chen & Sun, 2017; Zhao et al., 2022). For example, Meijers et al. (2016) discovered that European cities with more than 500,000 inhabitants benefit from the population agglomeration of adjacent areas, whereas cities with fewer than 250,000 inhabitants experience the negative effects of agglomeration shadow. In China, Wang and Tong (2017) demonstrated that there existed a substantial agglomeration shadow phenomenon in the Beijing metropolitan area. In addition, Li et al. (2023) explored the general patterns of agglomeration shadows in China, discovering that excessive development of the service sector in large national central cities leads to negative population growth in surrounding small and medium-sized cities. Additionally, the mutual borrowed effect of spatial spillover between two highly agglomerated cities must be considered. For example, Li, Li, et al. (2017) and Li, Zheng, and Liang (2017) demonstrated that cities with larger populations, such as Guangzhou and Foshan, could facilitate the flow of talent and materials between cities by constructing roads and bridges and trading commodities, which boosted the economic growth of both cities.

Therefore, we constructed conceptual diagrams that consider both intracity and intercity agglomeration (Fig. 1–b). The two main effects of intercity agglomeration economies are borrowed size (blue solid arrow) and agglomeration shadow (blue dashed arrow), and possible mutual borrowed (blue solid double-arrowed line) or mutual inhibition effects (gray dashed double-arrowed line). The blue dashed hexagon represents both the effect of intercity and intracity agglomeration economies, indicating that a high degree of agglomeration of neighboring cities positively promotes a city's economic growth. That is, the borrowed size effect is greater than the agglomeration shadow effect. Diagrammatically, there may exist multiple forces between two cities, to prevent visual clutter, we present a limited number of agglomeration relationships in Fig. 1(b). In addition, the perimeter of the local city is supposed to be six cities, but to simplify the problem, we only list two cities in the perimeter.

3.2. Theoretical formulas

Ciccone (2002) was the first to construct a model of the impact of population agglomeration on urban GDP per capita. This model is $\log(Q_s/N_s) = \theta(\log N_s/C_s) + \phi(\log K_s/N_s) + \varphi(\text{region}) + u_s$, where Q_s , N_s , C_s , K_s , and H_s are the total output, total labor force size, total area, capital stock, and human capital level of region s , respectively, and θ , ϕ , and φ are the coefficients for population density, capital stock, and human capital, respectively, $D(\text{region})$ is a region dummy variable, and u_s is the residual term. On this basis, Ciccone also constructed an empirical formula that describes the level of local urban economic growth in city s depends on the production density in neighboring city n . The formula is

$\log(Q_s/N_s) = \theta(\log N_s/C_s) + \phi(\log K_s/N_s) + \varphi \log H_s + D(\text{region}) + \omega(\log Q_{sn}/C_{sn}) + v_s$, where Q_{sn} and A_{sn} denote the total output and total area of the neighboring cities, respectively, ω denotes the coefficients of the productivity of the neighboring cities, and v_s is the residual term. On the other hand, population density and GDP per capita are endogenous to each other (Su et al., 2017). Considering that both urban population and economic activities are dependent on built-up areas, we draw on the idea of Chen, Li, et al. (2018) and Chen, Liu, et al. (2018), based on the Cobb–Douglas production function $Y_{it} = A_{it}K_{it}^\alpha L_{it}^{1-\alpha}$, where Y_{it} , A_{it} , K_{it} , and L_{it} denote the level of economic output, technological progress, capital and labor force of region i in the year t , respectively. Then,

simultaneously divide the exogenous built-up area variable on both sides of the equation and take the logarithm. That formula is $\ln(Y_{it}/C_{it}) = \ln A_{it} + (1 - \alpha)\ln(L_{it}/C_{it}) + \alpha \ln(K_{it}/C_{it})$ where $1 - \alpha$ is the coefficient of population density $\ln(L_{it}/C_{it})$ on per capita urban land revenue $\ln(Y_{it}/C_{it})$. In addition, we need comprehensive indicators to measure agglomeration economy. Therefore, we further draw on the comprehensive model proposed by Melo et al. (2009) to systematically measure the effect of agglomeration economy on urban economic growth. The model can be written as $Y_{it} = G_{it}(\cdot)F_{it}(X_{it})$, where $G_{it}(\cdot)$ includes a range of variables characterizing agglomeration economic indicator. Based on the existing empirical studies on measuring urban overall agglomeration (Abhishek et al., 2017; Cheng et al., 2020; Ciccone & Hall, 1996; Duranton, 2016; Glaeser, 2011), we selected population density, initial city size, and the dummy variable of whether it belongs to an urban agglomeration (If a city is located in the Beijing–Tianjin–Hebei, Yangtze River Delta, or Pearl River Delta urban agglomerations is equal to 1 and if otherwise equal to 0) to comprehensively measure the level of agglomeration economy.

The validation formula for Hypothesis 1.

Assuming that the closed city i has an income level Q_{it} , a population size N_{it} , and a built-up area C_{it} in year t . Then, based on the Cobb–Douglas production function, the formulas for the effect of agglomeration economy $AE(\text{Agglomeration Economy})$ on the income level of the city are as follows:

$$\ln\left(\frac{Q_{it}}{C_{it}}\right) = \alpha \ln AE_{it} + \beta n \left(\frac{K_{it}}{C_{it}}\right) + \gamma \ln H_{it} + \varphi X_{it} + \varepsilon_{it} \quad (1)$$

$$\ln\left(\frac{Q_{it}}{N_{it}}\right) = \alpha' \ln AE_{it} + \beta' n \left(\frac{K_{it}}{C_{it}}\right) + \gamma' \ln H_{it} + \varphi' X_{it} + \varepsilon'_{it} \quad (2)$$

In Eq. (1), Q_{it}/C_{it} is the per capita urban land revenue of city i in year t (the explained variable); AE_{it} is the level of agglomeration economy (the core explanatory variable); and α is the coefficients of the indicators of agglomeration economy AE_{it} .

For control variables, β , γ , and φ are the coefficients of the per capita urban land capital stock K_{it}/C_{it} , the level of human capital (number of patent applications per 10,000 persons) H_{it} , and all other control variables X_{it} , respectively, and ε_{it} is a randomized perturbation term. Eq. (2) is identical to Eq. (1), except that the explained variable is the GDP per capita of city i in year t , which is denoted by Q_{it}/N_{it} , and the core control variable is the capital stock per capita, which is denoted by K_{it}/N_{it} . To solve the aforementioned endogeneity problem, we also construct the first-order dynamic distributional lag model of the effect of agglomeration economy on urban economic growth.

The validation formula for Hypothesis 2.

First, the ideas of both borrowed size and agglomeration shadow suggest that there exist spatial spillover effects of population agglomeration. That is, a high degree of agglomeration in neighboring cities may promote or hinder the economic growth of the local city. Therefore, we draw on the spatial weights idea and adopt the spatial autoregressive model, spatial error model, and spatial Durbin model to verify whether urban population agglomeration has a spatial spillover effect. These models are presented as follows:

$$\ln\left(\frac{Q_{it}}{C_{it}} \text{ or } \frac{Q_{it}}{N_{it}}\right) = \alpha \ln AE_{it} + \rho W \ln\left(\frac{Q_{it}}{C_{it}} \text{ or } \frac{Q_{it}}{N_{it}}\right) + \beta \ln\left(\frac{K_{it}}{C_{it}} \text{ or } \frac{K_{it}}{N_{it}}\right) + \gamma \ln H_{it} + \varphi X_{it} + \varepsilon_{it} \quad (3)$$

$$\ln\left(\frac{Q_{it}}{C_{it}} \text{ or } \frac{Q_{it}}{N_{it}}\right) = \alpha \ln AE_{it} + \beta \ln\left(\frac{K_{it}}{C_{it}} \text{ or } \frac{K_{it}}{N_{it}}\right) + \gamma \ln H_{it} + \varphi X_{it} + u_{it}, \quad u_{it} = \lambda W \varepsilon_{it} + \varepsilon_{it} \quad (4)$$

$$\ln\left(\frac{Q_{it}}{C_{it}} \text{ or } \frac{Q_{it}}{N_{it}}\right) = \alpha \ln AE_{it} + \beta \ln\left(\frac{K_{it}}{C_{it}} \text{ or } \frac{K_{it}}{N_{it}}\right) + \gamma \ln H_{it} + \varphi X_{it} + \rho W \ln\left(\frac{Q_{it}}{C_{it}} \text{ or } \frac{Q_{it}}{N_{it}}\right) + \alpha' W \ln AE_{in,t} + \beta' W \ln\left(\frac{K_{in,t}}{C_{in,t}} \text{ or } \frac{K_{in,t}}{N_{in,t}}\right) + \gamma' W \ln H_{in,t} + \varphi' W X_{in,t} + \varepsilon_{it} \quad (5)$$

Eq. (3) is a spatial autoregressive model describing the spatial spillover effects on a city's economic growth from neighboring cities, ρ is the coefficient of the spatial lag term. Eq. (4) is a spatial error model describing the effect on a city's economic growth resulting from neighboring cities through the spatial dependence of the spatial error term, λ is the coefficient of the spatial error term. Eq. (5) is a spatial Durbin model that focuses on the impact of the agglomeration of neighboring cities on a city's economic growth, in this equation, α' , β' , γ' , and φ' are the coefficients for population density, per capita urban land capital stock or capital stock per capita, human capital, and other control variables (neighboring city) on a city's economic growth, respectively.

More importantly, we add the dummy variable “proximity to high-tier cities” $D(Hsize)_{in,t}$ and the interaction term between the dummy variable and local population agglomeration $AE_{it} \times D(Hsize)_{in,t}$, based on the foundation regressions (1) and (2). In particular, $D(Hsize)_{in,t}$ represents the indicator of population agglomeration in neighboring cities (a city is equal to 1 if and only if the city is adjacent to a city that is in the top 5 % of cities in the country by population, and vice versa as 0). Specifically, Eq. (6) verifies whether the impact of the population agglomeration of neighboring cities on the economic growth of local cities is facilitating (borrowed size) or inhibiting (agglomeration shadow). Eq. (7) verifies whether such proximity strengthens or weakens the contribution of the intracity agglomeration economy to the economic growth of the local city.

$$\ln\left(\frac{Q_{it}}{C_{it}} \text{ or } \frac{Q_{it}}{N_{it}}\right) = \alpha \ln AE_{it} + \beta \ln\left(\frac{K_{it}}{C_{it}} \text{ or } \frac{K_{it}}{N_{it}}\right) + \gamma \ln H_{it} + \kappa D(Hsize)_{in,t} + \varphi X_{it} + \varepsilon_{it} \quad (6)$$

$$\ln\left(\frac{Q_{it}}{C_{it}} \text{ or } \frac{Q_{it}}{N_{it}}\right) = \alpha \ln AE_{it} + \beta \ln\left(\frac{K_{it}}{C_{it}} \text{ or } \frac{K_{it}}{N_{it}}\right) + \gamma \ln H_{it} + \eta D(Hsize)_{in,t} * \ln AE_{it} + \varphi X_{it} + \varepsilon_{it} \quad (7)$$

In Eqs. (6) and (7), κ is the coefficient of proximity to high-tier cities, and η is the coefficient of the interaction term of proximity to a high-tier city with the three indicators of agglomeration economy.

3.3. Data sources

We used 285 prefecture-level cities in China from 2005 to 2020 as an example and selected per capita urban land revenue and GDP per capita to measure the level of urban economy (the explained variable). GDP data were obtained from the 2006–2021 *China City Statistical Yearbook*. However, the 2017 *China City Statistical Yearbook* only included the GDP data of a city's districts rather than that of the whole city. Thus, we obtained data on city-level GDP in 2017 from the statistical bulletins of each of the 285 prefectural-level cities. We transformed nominal GDP into real GDP, and data for determining the GDP deflator were obtained from the 2006–2021 *China Statistical Yearbook*. By using the more credible Chinese census data for 2010 and 2020 as a benchmark (China's population census is held every 10 years), we obtained data on population size (specifically, on the number of permanent residents) from the statistical yearbooks (more than 500 yearbooks) of each of the 35 provinces for each year from 2006 to 2021 and verified them against the statistical bulletins (4560) of each city. The provincial statistical yearbooks for Liaoning, Jilin, and Heilongjiang only contained data on the household population. Thus, we calculated the number of permanent residents by adding the household population figures provided in the *China City Statistical Yearbook* with the temporary resident population

figures provided in the *China Urban Construction Statistical Yearbook*.

Data on a city's built-up area were obtained using DMSP/OLS and NPP/VIIRS nighttime light imagery data (<http://ngdc.noaa.gov/eog>) because official statistics on this variable are scarce. Despite the overlapping periods of the DMSP/OLS (2005–2012) and NPP/VIIRS (2013–2020) datasets, covering both these periods (Elvidge et al., 2021), these two datasets differ in spatial resolution, spectral response, the point spread function of the sensors, and the range of the DNB radiance (Li, Li, et al., 2017). Therefore, by using the method of Chen et al. (2019), we performed heterologous correction in four steps as follows. First, raw monthly NPP/VIIRS images for 2013–2020 were subjected to data preprocessing through annual mean synthesis, resampling to 0.008333°, and low-brightness rejection. Second, the luminance of the preprocessed NPP/VIIRS image was transformed using the power function equation $Y = a * X^b * M_{\theta, \sigma} + c$. In this equation, X and Y represent image gray value matrices; a , b , and c are the parameters to be solved; M is a Gaussian low-pass filter; θ and σ denote the filter size and the standard deviation of the Gaussian distribution, respectively; and $*$ denotes the spatial convolution. The size of the Gaussian low-pass filter was set to 13. Third, the luminance saturation effect of DMSP/OLS was simulated for the power function-transformed NPP/VIIRS on the basis of the formula $y'_{ij} = \begin{cases} y_{ij} & y_{ij} \leq c \\ c & y_{ij} > c \end{cases}$, where y_{ij} and y'_{ij} represent the gray values of the i th row and the j th column in matrix Y before and after saturation correction, respectively, and the luminance saturation threshold c was set to 50. Fourth, mask data were produced using data on watershed land from the 2015 China Land Use and Cover Change data and the Gas Flare product to remove nonurban light sources from the nighttime lighting dataset. Furthermore, we applied the temporal and spatial normalization model to extract data on built-up area in each year (Yao et al., 2018). Specifically, we used 30-m data from the Chinese Land Use and Land Cover Change dataset in 2010 and 2015 as the baseline, and we split the national light data into 2800 NTL images based on district and county boundaries. Subsequently, the formula $|Area_{t,z} - S| = \min$ was applied, where $Area_{t,z}$ and S denote the urban construction land and the baseline built-up area estimated for year t at threshold z , respectively, and to obtain the optimal thresholds for each city in each year, the results under different thresholds were tested against the baseline built-up area data by using the exhaustive enumeration method. On the basis of this threshold, the heterogeneous corrected DMSP/OLS and NPP/VIIRS data were extracted to obtain data on the construction land of 285 cities in consecutive years (Wei et al., 2014; Fig. 2).

The control variables include per capita urban land capital stock, capital stock per capita, share of manufacturing employment, number of patent applications per 10,000 persons, and average annual temperature. Real capital stock was calculated using the perpetual inventory method, and a uniform of depreciation rate of 9.6 % was applied (Chen, Li, et al., 2018). Data on the number of patent applications per 10,000 persons, which is an indicator of human capital, were collected from the statistical bulletins of each city in the sample and from the *China National Intellectual Property Administration* (<https://www.cnipa.gov.cn>); these data were corroborated against those from the statistical yearbooks and science and technology yearbooks of the 35 provinces in the sample. Data were drawn from these sources because the *China City Statistical Yearbook* only includes patent data from 2017 onwards. Data on the share of manufacturing employment were obtained from the 2006–2021 *China City Statistical Yearbook* and were supplemented by data from selected parts of the *Tabulation On 2020 China Population Census by County*. Average temperature data were obtained from a private company, namely *Shanghai Bancheng Data Technology Company* (<https://www.lifangshuju.com>), because temperature data for prefecture-level cities are absent in the *China Statistical Yearbook*, *China City Statistical Yearbook*, and *China Statistical Yearbook On Environment*. These temperature data were reliable: we corroborated the temperature data by comparing 10 % of randomly selected data points against the average annual temperature data of major cities in the *China Statistical*

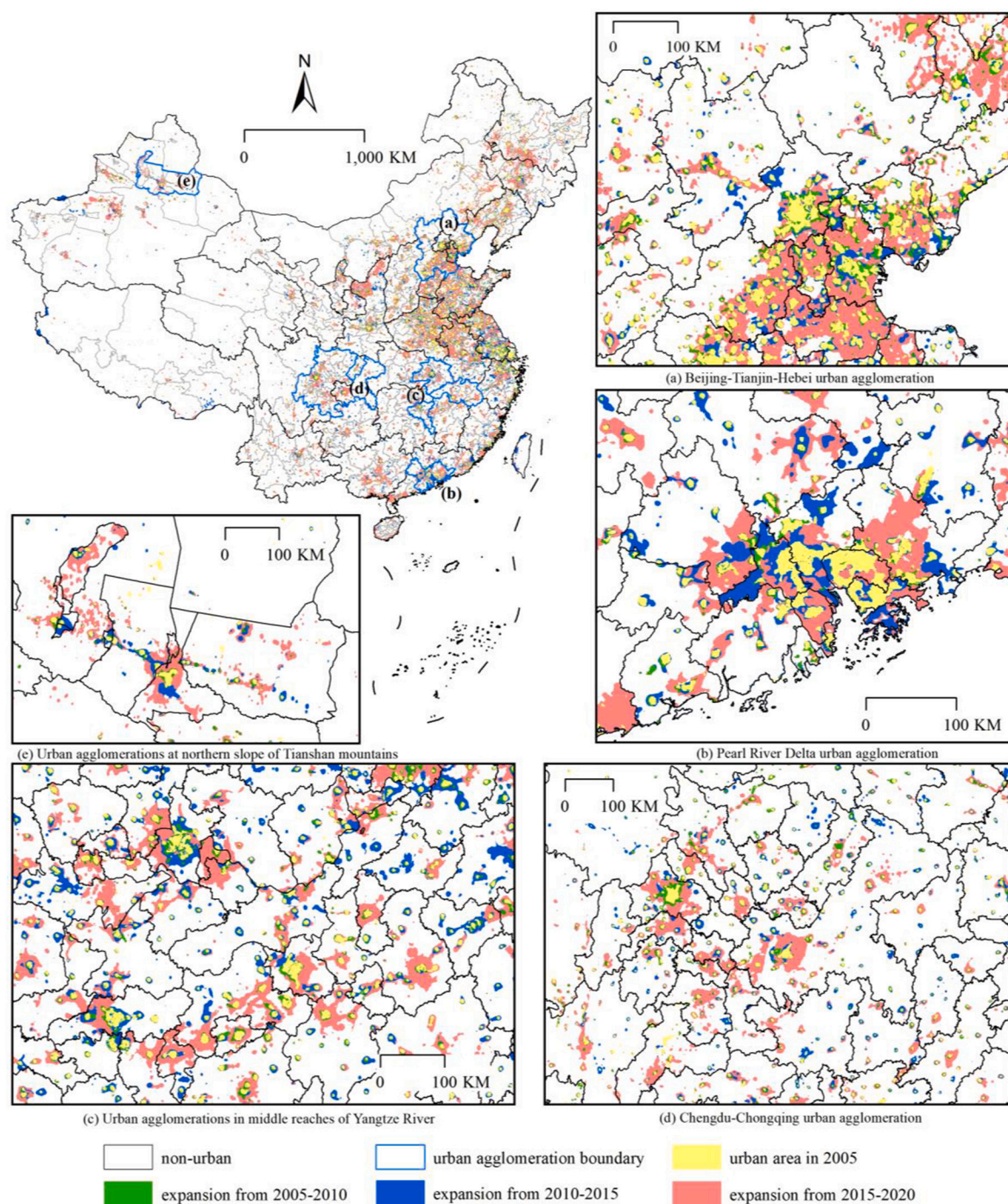


Fig. 2. Built-up area of 285 prefectural-level cities in China, 2005–2020.

Yearbook On Environment, and the error was small at 2°C – 1.2°C .

The instrumental variables (IVs) that were assessed comprise the relief degree of land surface (RDLS) and river network density for population density and slope for initial city size. The RDLS and slope data were extracted from 30-m digital elevation model data from the Geo-spatial Data Cloud (<https://www.gscloud.cn/>), and the river network density data were obtained from the Chinese Scientific Database (<http://www.csdata.org>). Additionally, we modified the macrostatistical GDP data with nighttime luminance data through the following steps. First, we derived the nighttime luminance values for 285 Chinese cities from 2005 to 2020 from the study by Wu et al. (2021), which was included in the Harvard University research database. Second, using the methodology of Chen, Li, et al. (2018) and Chen, Liu, et al. (2018), we

adopted the formula $\ln(\text{GDP}_i) = a\ln(\text{DN}_i)^2 + b\ln(\text{DN}_i) + c$ to correct the GDP data from Chinese statistical yearbooks year by year on the basis of the luminance values, where a , b , and c are the coefficients of the square, liner term, and intercept terms of the nighttime luminance values DN for city i , respectively. The descriptive statistics of the variables are presented in Table 1.

4. Results

4.1. Regression results for intracity agglomeration

Table 2 shown the results of panel regression and GMM estimation by using Eqs. (1) and (2) to analyze the effect of population agglomeration

Table 1
Descriptive statistics of variables.

	Sample	Mean value	Standard error	Minimum value	P50	Maximum value
Ln(per capita urban land revenue)	4560	8.511	0.863	5.123	8.562	11.613
Ln(GDP per capita)	4560	9.582	0.559	7.792	9.533	11.79
Ln(population density)	4560	8.14	0.802	5.032	8.122	11.313
Ln(initial city size)	4560	5.825	0.686	2.899	5.872	7.937
D(belong to an urban agglomeration)	4560	0.119	0.324	0	0	1
Ln(per capita urban land capital stocks)	4560	10.255	0.78	6.752	10.303	13.063
Ln(capital stock per capita)	4560	11.326	0.907	8.307	11.441	14.095
Patent applications per 10,000 persons	4560	11.501	18.978	0.007	4.353	176.654
Share of manufacturing employment	4560	0.238	0.137	0.006	0.213	0.813
Average annual temperature	4560	14.613	5.14	−1.21	15.607	26.441
D(proximity high-tier cities)	4560	0.263	0.44	0	0	1
Ln(modified per capita urban land revenue)	4560	8.511	0.863	5.093	8.556	11.506
Ln(modified GDP per capita)	4560	9.582	0.557	7.776	9.531	11.790
Ln(relief degree of land surface)	4560	6.854	1.659	3.296	6.250	9.593
River network density	4560	0.430	0.475	0.011	0.304	4.369
Slope	4560	9.621	6.158	1.183	9.096	27.437

on Chinese urban economic growth from 2005 to 2020. The results indicated that intracity population agglomeration significantly and positively influenced urban economic growth in China. The effects of population density and initial city size on per capita urban land revenue had coefficients that were stable at approximately 0.87 and 0.03, respectively; however, the effects of these two indicators were not robust about GDP per capita.

The explained variable in Models 1–8 was per capita urban land revenue. Models 1 and 2 used population density to represent population agglomeration. An individual fixed-effects model was chosen because $p = 0.000$ was obtained in a Hausman test. According to the results, population density had a positive effect on per capita urban land revenue (coefficients = 0.87–0.9, significant at the 1 % level), indicating that intracity agglomeration promotes urban economic growth in China. In general, every 10 % increase in urban population density was associated with an 8.7 % increase in per capita urban land revenue. This result was close to the findings of [Ciccone and Hall \(1996\)](#) and [Ciccone \(2002\)](#), who concluded that the effect of population density on the economic growth of cities in the United States and Europe was 6 % and 4.5 %, respectively. Models 3 and 4 used initial city size to represent population agglomeration. Time and individual double-fixed-effects models were chosen because the initial city size is static over time. The results shown that a doubling of the initial population size of the city would increase the per capita urban land revenue by 2.9 %–4 %. This result was consistent with [Abhishek et al. \(2017\)](#) who noted that initial population size significantly promoted urban economic growth in India. Model 5 adopted the dummy variable of belong to an urban agglomeration indicated agglomeration economy. The results indicated that joining an urban agglomeration with strong regional network linkages also can boost urban economic growth, with an average degree of increase of 0.65, compared with cities not belonging to the urban agglomeration. As stated in the Methods section, dynamic panel models with a first-order lag term of income level (Models 6–8) were estimated using GMM to address endogeneity problems. The results were similar to the results of previous models (Models 1–5), population density, initial city size, and the dummy variable of belong to an urban agglomeration were positively associated with per capita urban land revenue. This agreement between models indicated the robustness of our results and the veracity of our models.

The explained variable in Models 9–16 was GDP per capita, and population density, initial city size, and the dummy variable of belong to an urban agglomeration were indicators of agglomeration economy in Models 9–13. According to the results and similar to the results of the aforementioned models, these three indicators had a significant and positive effect on GDP per capita. However, the effects of these three indicators on GDP per capita in Models 14–16 (GMM-estimated results) tended to be weaker, be nonsignificant, and had opposite signs relative

to those in Models 9–13. For example, the effect of population density had a coefficient of 0.04 in Model 9 but 0.002 in Model 14, the effect of initial city size was nonsignificant in Model 15, and the effect of the dummy variable of belong to an urban agglomeration had opposite signs in Model 13 and 16. This disagreement suggested that the regression results were less robust when GDP per capita, as opposed to per capita urban land revenue, was the explained variable. The possible reason was that the existence of about 40 % of resource-based cities in China, characterized by rapid wealth growth and urban construction, and the level of population agglomeration was slower. From this, we assumed that the wealth of resource-based cities was driven by land urbanization rather than population urbanization, in which population density significantly affected per capita urban land revenue rather than GDP per capita in Chinese resource-based cities, and initial population size was not closely related to economic growth in these cities. Therefore, we specifically explored the impact of population agglomeration on urban economic growth in 115 resource-based cities in China from 2005 to 2020. The results showed that the influence coefficient of population density on per capita urban land revenue was stable at 0.87, while its effect on GDP per capita was not significant at the 1 % level. Even worse, the initial city size had a negative impact on both explained variables.

Regarding the control variables, per capita urban land capital stock had a positive effect on per capita urban land revenue. The coefficient of per capita urban land capital stock in Models 6–8 was stable at approximately 0.19; this finding indicated the importance of capital investment for China's economic growth even as China is settling into its middle income status. Share of manufacturing employment also had a positive effect on per capita urban land revenue; this result indicated that manufacturing agglomeration still plays a key role in the economic growth of Chinese cities. However, results for the capital stock per capita and the share of manufacturing employment were not corroborated by the results for effects on GDP per capita, indicating a lack of robustness. The number of patent applications per 10,000 persons had coefficients of −0.003 to 0.007 relative to per capita urban land revenue; this finding contradicts the conventional wisdom that innovation promotes economic growth ([Raspe & Oort, 2006](#)). This may be because China does not yet have a sufficiently mature economy for innovation to promote growth. The results of the effect of the average annual temperature on urban income were also not robust. Specifically, the coefficients signs of fixed-effects were contrary to GMM estimation. Thus, we added the quadratic term for the average annual temperature to examine whether a nonlinear relationship is present; the results suggested an inverted U-shaped relationship between the average annual temperature and economic growth (coefficient of quadratic term = −0.07, significant at 1 % level).

Taken together, these results suggest the positive influence of intracity population agglomeration (for all three indicators) on urban

Table 2

Panel regression and GMM estimation results on effect of agglomeration economy on economic growth.

Explained variable: per capita urban land revenue	Population density		Initial city size		Urban agglomeration	GMM estimation		
	Model 1 (FE)	Model 2 (FE)	Model 3 (xi:reg)	Model 4 (xi:reg)	Model 5 (xi:reg)	Model 6 (GMM)	Model 7 (GMM)	Model 8 (GMM)
Ln(per capita urban land revenue)t-1						0.549*** (59.896)	0.965*** (256.013)	0.958*** (247.494)
Ln(population density)	0.897*** (233.773)	0.869*** (229.257)				0.473*** (57.676)		
Ln(initial city size)			0.029** (2.339)	0.040*** (2.830)			0.031*** (5.726)	
D(belong to an urban agglomeration)					0.649*** (5.356)			0.060*** (4.164)
Ln(per capita urban land capital stocks)	0.142*** (30.762)	0.144*** (33.165)	0.808*** (110.931)	0.808*** (111.104)	0.808*** (111.306)	0.190*** (45.473)	0.197*** (61.191)	0.197*** (60.034)
Patent applications per 10,000 persons	−0.003*** (−13.301)	−0.001*** (−8.041)	0.006*** (26.204)	0.007*** (26.537)	0.007*** (26.575)	0.003*** (13.430)	−0.003*** (−13.407)	−0.003*** (−13.343)
Share of manufacturing employment		0.490*** (13.768)		0.204*** (4.392)	0.204*** (4.398)	0.756*** (17.407)	0.132*** (4.450)	0.112*** (3.448)
Average annual temperature		−0.084*** (−19.344)		0.014* (1.819)	0.014* (1.779)	−0.021*** (−16.877)	−0.009*** (−12.308)	−0.008*** (−11.112)
Constant	−0.223*** (−4.060)	1.086*** (12.582)	1.338*** (13.840)	1.077*** (9.379)	0.726*** (3.475)	−1.917*** (−40.143)	−1.869*** (−34.219)	−1.648*** (−37.612)
Wald test						3.37e+06***	9.82e+06***	1.10e+07***
Hausman test p-value	0.000	0.000						
R ²	0.943	0.950	0.971	0.971	0.971			
N	4560	4560	4544	4544	4560	4275	4260	4275

Explained variable: GDP per capita	Population density		Initial city size		Urban agglomeration	GMM estimation		
	Model 9 (FE)	Model 10 (FE)	Model 11 (xi:reg)	Model 12 (xi:reg)	Model 13 (xi:reg)	Model 14 (GMM)	Model 15 (GMM)	Model 16 (GMM)
Ln(GDP per capita)t-1						1.012*** (415.925)	1.013*** (418.196)	1.016*** (403.993)
Ln(population density)	0.040*** (6.833)	0.013** (2.356)				0.002** (2.222)		
Ln(initial city size)			0.033*** (3.883)	0.072*** (7.499)			−0.001 (−0.980)	
D(belong to an urban agglomeration)					0.524*** (6.317)			−0.010*** (−3.303)
Ln(capital stock per capita)	0.142*** (30.762)	0.144*** (33.165)	0.388*** (49.457)	0.382*** (49.148)	0.382*** (49.189)	−0.029*** (−23.333)	−0.030*** (−33.767)	−0.031*** (−33.735)
Patent applications per 10,000 persons	−0.003*** (−13.301)	−0.001*** (−8.041)	0.002*** (9.383)	0.002*** (10.472)	0.002*** (10.486)	0.000 (0.365)	0.000 (0.525)	0.000 (1.606)
Share of manufacturing employment		0.490*** (13.768)		0.303*** (9.520)	0.303*** (9.528)	−0.042*** (−5.054)	−0.041*** (−4.849)	−0.041*** (−4.971)
Average annual temperature		−0.084*** (−19.344)		−0.015*** (−2.886)	−0.015*** (−2.936)	0.001*** (3.464)	0.001*** (5.172)	0.001*** (5.316)
Constant	7.676*** (82.545)	8.968*** (81.379)	6.041*** (63.118)	5.922*** (56.419)	5.933*** (36.812)	0.203*** (9.009)	0.238*** (10.659)	0.216*** (10.770)
Wald test						3.74e+08***	3.69e+08***	3.55e+08***
Hausman test p-value	0.000	0.000						
R ²	0.271	0.359	0.967	0.968	0.967			
N	4560	4560	4544	4544	4560	4275	4260	4275

Note: *t* statistics in parentheses; ***, **, and * indicate significance at the 1 %, 5 %, and 10 % significance levels, respectively. Because the permanent resident population of Chongqing city was outlier, the data of Chongqing city were excluded from the model with initial city size, for which the sample size was 4544. The sample size in the GMM test was 4260.

economic growth. The results were robust for per capita urban land revenue, but not GDP per capita.

We further determined whether factors related to location (eastern, middle, or western region of China), policy (provincial capital, planned city, or coastal open city where the government played an important role in its economic growth in the early days of reform and opening up), and the natural environment (tropical or temperate climate)—all of which were dummy variables—moderates the effect of agglomeration economy on economic growth (Table 3). This positive effect (specifically, the positive effect of population density on economic growth) was stronger in the eastern and central than in the western parts of China. This positive effect was present but weak in provincial capitals and open coastal cities, but the results were less robust than the results for location.

Specifically, Models 1–4 included the interaction terms of population

density with location, policy, or natural environment. The result of Model 1 shown that population agglomeration contributes more to economic growth in eastern and central cities than in western cities (the interaction coefficient of east and middle are 0.019 and 0.032, respectively). In Model 2, the coefficient of the interaction term was 0.023; it was nonsignificant at the 10 % level, but its *t* value was 1.62. This result indicated that the agglomeration–growth effect was stronger in provincial capitals and planned cities than in other cities. The coefficient of Model 3 was nonsignificant, which indicated that early market policies had a weak effect on the economic growth of Chinese cities. The results of Model 4 indicated that the coefficient of the agglomeration–growth effect was 0.125 higher in temperate cities than in tropical cities. This finding might be because Chinese cities with a temperate climate were generally sparsely populated, and the spillover effects were slight. By

Table 3

Spatial differences in the effect of agglomeration economy on economic growth.

Core explanatory variables: population density	Location	Policy		Natural factor	Location	Policy		Natural factor
	Model 1 (FE)	Model 2 (FE)	Model 3 (FE)	Model 4 (FE)	Model 5 (GMM)	Model 6 (GMM)	Model 7 (GMM)	Model 8 (GMM)
Ln(population density)	0.852*** (148.853)	0.868*** (228.248)	0.869*** (228.831)	0.825*** (190.444)	0.491*** (60.473)	0.489*** (58.061)	0.480*** (57.383)	0.480*** (57.242)
Ln(population density) × D(East)	0.019** (2.257)				0.027*** (12.549)			
Ln(population density) × D(Middle)	0.032*** (4.125)				0.011*** (5.776)			
Ln(population density) × D(provincial capitals and planned cities)		0.023 (1.615)				0.015*** (5.582)		
Ln(population density) × D(open coastal cities)			−0.016 (−0.915)				0.015*** (6.658)	
Ln(population density) × D(temperate zone)				0.125*** (18.934)				0.008*** (3.999)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Hausman test p-value	0.000	0.000	0.000	0.000				
Wald test					3.67e+06***	3.03e+06***	3.04e+06***	3.37e+06***
R ²	0.95	0.95	0.95	0.954				
N	4560	4560	4560	4560	4275	4275	4275	4275

Core explanatory variables: initial city size	Location	Policy		Natural factor	Location	Policy		Natural factor
	Model 9 (xi:reg)	Model 10 (xi:reg)	Model 11 (xi:reg)	Model 12 (xi:reg)	Model 13 (GMM)	Model 14 (GMM)	Model 15 (GMM)	Model 16 (GMM)
Ln(initial city size)	−0.082*** (−4.222)	−0.140*** (−5.961)	0.040*** (2.83)	0.054*** (3.43)	0.018*** (3.156)	0.045*** (8.785)	0.033*** (6.298)	0.037*** (6.922)
Ln(initial city size) × D(East)	0.074*** (7.868)				0.010*** (5.499)			
Ln(initial city size) × D(Middle)	0.036** (2.393)				0.009*** (5.747)			
Ln(initial city size) × D(provincial capitals and planned cities)		0.109*** (9.104)				−0.015*** (−7.348)		
Ln(initial city size) × D(open coastal cities)			0.014 (1.421)				−0.006** (−2.265)	
Ln(initial city size) × D(temperate zone)				−0.014 (−1.421)				−0.007*** (−3.553)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Wald test					9.69e+06***	9.95e+06***	9.54e+06***	9.96e+06***
R ²	0.971	0.971	0.971	0.971				
N	4544	4544	4544	4544	4260	4260	4260	4260

Note: *t* values are in parentheses in the FE and xi:reg models, and *z* values are in parentheses in the GMM model; ***, **, and * indicate significance at the 1 %, 5 %, and 10 % significance levels, respectively. The explained variable was per capita urban land revenue. The control variables were per capita urban land capital stock, number of patent applications per 10,000 persons, share of manufacturing employment, and average annual temperature.

contrast, cities with a tropical climate in China were more attractive and densely populated, because good quality-of-life amenities and well-developed transportation networks tend to attract population agglomeration (Yang et al., 2019). However, the cost of living was also higher, so nominal incomes were not significantly higher. Models 9–12 describe the effects of the interaction terms of initial city size with location, policy, or natural factor on per capita urban land revenue. Similarly, Model 9 indicated that the agglomeration–growth effect was stronger in the eastern and central regions of the country than in the western regions of the country. However, the results for the interaction term of initial city size was not robust, as indicated by the negative coefficients of the effect of initial city size on per capita urban land revenue in Models 9 and 10. To solve the aforementioned endogeneity problem, we used the GMM dynamic panel estimation to corroborate the aforementioned results (see Models 5–8 and 13–16). The results demonstrated that the spatial difference in the effect of agglomeration economy on the economic growth of Chinese cities was reasonable.

Moreover, we selected GDP per capita as the explained variable to verify spatial differences in agglomeration economy (Table 4), and the results further demonstrated that location and policy moderated the agglomeration–growth effect.

In addition, the results of Tables 3 and 4 similarly indicated that the results were not robust when using GDP per capita to measure economic growth, specifically focused on the western cities. The possible reasons for the results were still the negative effect of resource-based cities. Specifically, first, the resource-based cities in the West account for 51 % of the total. Second, the large amount of undeveloped land in the western region facilitated land urbanization faster than population urbanization, leading to a drop in population density while GDP per capita rose. For example, in Yulin City, located in the northwest, the built-up area increased by 92.2 % from 2005 to 2020 due to the construction of high-tech industrial parks and science and innovation cities, and the permanent population only increased by 9.87 %. Although its population density was decreasing, real GDP per capita increased by 152.7 % during the same period. Another example was the city of Yuxi in southwest China, whose built-up area increased from 265 km² in 2005 to 992.5 km² in 2020, while its permanent population increased by only 1.6 %. As a result, its population density decreased by 79 %, while its real GDP per capita increased by 27.9 % over the same period.

In summary, positive intracity agglomeration effects on economic growth were stronger in eastern and central cities than in western cities. This positive effect was also stronger in provincial capitals and planned

Table 4
Spatial differences in the effect of agglomeration economy on economic growth.

Core explanatory variables: population density	Location	Policy		Natural factor	Core explanatory variables: initial city size	Location	Policy		Natural factor
	Model 1 (FE)	Model 2 (FE)	Model 3 (FE)	Model 4 (FE)		Model 5 (xi:reg)	Model 6 (xi:reg)	Model 7 (xi:reg)	Model 8 (xi:reg)
Ln(population density)	−0.005 (−0.661)	0.012** (2.196)	0.014** (2.432)	−0.032*** (−5.415)	Ln(initial city size)	−0.211*** (−8.708)	−0.202*** (−13.903)	0.072*** (7.499)	0.057*** (4.714)
Ln(population density)× D(East)	0.019** (2.257)				Ln(initial city size)× D(east)	0.171*** (13.64)			
Ln(population density)× D(Middle)	0.032*** (4.125)				Ln(initial city size)× D(middle)	0.128*** (15.936)			
Ln(population density)× D(provincial capitals and planned cities)		0.023 (1.615)			Ln(initial city size)× D(provincial capitals and planned cities)		0.166*** (17.444)		
Ln(population density)× D(open coastal cities)			−0.016 (−0.915)		Ln(initial city size)× D(open coastal cities)			0.018*** (2.741)	
Ln(population density)× D(temperate zone)				0.125*** (18.934)	Ln(initial city size)× D(temperate zone)				0.016** (2.041)
Control variables	Yes	Yes	Yes	Yes	Control variables	Yes	Yes	Yes	Yes
Hausman test p-value	0.000	0.000	0.000	0.000	Hausman test p-value	/	/	/	/
R ²	0.361	0.359	0.359	0.408	R ²	0.968	0.968	0.968	0.968
N	4560	4560	4560	4560	N	4544	4544	4544	4544

Note: *t* statistics in parentheses; ***, **, and * indicate significance at the 1 %, 5 %, and 10 % significance levels, respectively. The explained variable is GDP per capita. The control variables were capital stock per capita, number of patent applications per 10,000 persons, share of manufacturing employment, and average annual temperature. We also calculated GMM estimates of the effect of agglomeration economy on GDP per capita, the results of which were consistent with the fixed effects regression results, which we do not present here to avoid redundancy.

cities than in other cities. The difference in the strength of this effect had gradually narrowed between coastal and inland cities over time. Furthermore, the agglomeration–growth effect was not moderated by natural environment-related factors.

4.2. Regression results for intercity agglomeration

Based on Eqs. (3) to (5), we first verified whether urban population agglomeration had spatial spillover effects in China (Table 5). The results indicated that population agglomeration had significant positive

Table 5
Spatial spillover effects of economic growth.

	Explained variable: per capita urban land revenue			Explained variable: GDP per capita		
	Model 1 (SAR)	Model 2 (SEM)	Model 3 (SDM)	Model 4 (SAR)	Model 5 (SEM)	Model 6 (SDM)
Ln(population density)	0.757*** (131.225)	0.845*** (159.256)	0.767*** (96.604)	−0.004 (−0.773)	−0.013** (−2.350)	−0.029*** (−4.910)
Ln(per capita urban land capital stocks)	0.163*** (40.440)	0.142*** (25.624)	0.204*** (27.154)			
Ln(capital stock per capita)				0.081*** (19.793)	0.142*** (25.624)	0.204*** (28.376)
Patent applications per 10,000 persons	−0.001*** (−3.458)	−0.001*** (−5.330)	0.000 (0.305)	−0.001*** (−8.415)	−0.001*** (−5.330)	0.000 (0.301)
Share of manufacturing	0.359*** (10.968)	0.456*** (14.252)	0.426*** (13.574)	0.423*** (14.011)	0.456*** (14.252)	0.426*** (13.595)
Average annual temperature	−0.058*** (−14.196)	−0.076*** (−10.848)	−0.012 (−1.155)	−0.046*** (−11.925)	−0.076*** (−10.848)	−0.012 (−1.157)
W_Ln(population density)			0.106*** (9.666)			0.060*** (4.514)
W_Ln(per capita urban land capital stocks)			−0.047*** (−4.018)			
W_Ln(capital stock per capita)						−0.047*** (−4.408)
W_Patent applications per 10,000 persons			−0.002*** (−5.662)			−0.002*** (−5.596)
W_Share of manufacturing			0.006 (0.070)			0.003 (0.036)
W_Average annual temperature			−0.070*** (−5.130)			−0.069*** (−5.124)
ρ	0.178*** (24.220)		0.571*** (41.087)	0.488*** (34.828)		0.571*** (41.087)
λ		0.579*** (39.333)			0.579*** (39.333)	
Log L	3474.4146	3791.5157	3902.7659	3684.0343	3791.5157	3902.7659
R ²	0.481	0.514	0.539	0.042	0.063	0.099
N	4560	4560	4560	4560	4560	4560

Note: *z* statistics in parentheses; ***, **, and * indicate significance at the 1 %, 5 %, and 10 % significance levels, respectively. Models 3 and 6 report the direct and indirect effects of the spatial Durbin model, with the direct effect, such as Ln(population density), at the front and the indirect effect, such as W_Ln(population density), at the back. Agglomeration economy was only indicated by population density in the spatial panel regressions because initial city size and belongs to an urban agglomeration are time-independent indicators.

spatial spillover effects, but intracity agglomeration was more powerful than intercity agglomeration as a driver of economic growth. Second, the contribution of intracity population agglomeration to economic growth declined slightly after including intercity agglomeration elements. Furthermore, the level of economy in neighboring cities had a strong positive spatial spillover effect on a city's economic growth, which means there existed a spatial correlation of strong–strong association between cities.

Economic growth had a significant spatial correlation. Specifically, the global Moran's I values of per capita urban land revenue and GDP per capita ranged from 0.26 to 0.53 and were significant at the 1 % level. Furthermore, the results revealed that economic growth in a city was higher when its neighboring cities had larger economies. There existed a strong–strong union development pattern among cities. Specifically, according to the regression results of Models 1–3, the spatial lag coefficient ρ , spatial error coefficient λ , and spatial autocorrelation coefficient ρ were 0.178, 0.579 and 0.571, respectively, and were all significant at the 1 % level. The results of Models 3 and 6, which were

spatial Durbin models, included both direct and indirect effects. In Model 3, the coefficients of the direct and indirect effects of population density were 0.767 and 0.106 (all significant at the 1 % level), respectively. This result indicated that intracity population agglomeration has positive spatial spillover effects. In addition, this result also revealed that intracity agglomeration was more powerful than intercity agglomeration as a driver of economic growth. They contributed 88 % and 12 % to the effects of the overall agglomeration economy, respectively. Meanwhile, the coefficient of population density on per capita urban land revenue was still positive (0.757, 0.845, 0.767, all significant at 1 %) in Models 1–3, which revealed that intracity population agglomeration still significantly promoted urban economic growth when intercity population agglomeration effects were accounted for. However, there was a slight reduction in the contribution of intracity population agglomeration to economic growth, after incorporating elements of intercity agglomeration. Specifically, the coefficient of population density changes from 0.869 (Model 2 of Table 2) to 0.767 (Model 3 of Table 5), a decrease of about 0.1 (all significant at the 1 % level).

Table 6
Effects of intercity agglomeration on economic growth.

Explained variable: per capita urban land revenue	Population density		Initial city size		Urban agglomeration	
	Model 1 (xi:reg)	Model 2 (xi:reg)	Model 3 (xi:reg)	Model 4 (xi:reg)	Model 5 (xi:reg)	Model 6 (xi:reg)
Ln(population density)	0.592*** (70.301)	0.593*** (69.813)				
Ln(initial city size)			0.134** (2.310)	−0.021 (−1.263)		
D(belong to an urban agglomeration)					0.499*** (5.243)	0.427** (2.426)
D(proximity high-tier cities)	0.538*** (6.531)		0.201** (2.048)		0.068 (0.395)	
Ln(population density)× D(proximity high-tier cities)		−0.006 (−0.987)				
Ln(initial city size)× D(proximity high-tier cities)				0.037*** (3.282)		
D(belong to an urban agglomeration)× D(proximity high-tier cities)						−0.253** (−2.360)
Constant	0.610*** (4.290)	1.191*** (12.716)	0.186 (0.432)	1.252*** (9.000)	0.808*** (10.900)	1.202*** (18.030)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.987	0.987	0.971	0.971	0.971	0.971
N	4560	4560	4544	4544	4560	4560

Explained variable: GDP per capita	Population density		Initial city size		Urban agglomeration	
	Model 7 (xi:reg)	Model 8 (xi:reg)	Model 9 (xi:reg)	Model 10 (xi:reg)	Model 11 (xi:reg)	Model 12 (xi:reg)
Ln(population density)	−0.041*** (−7.530)	−0.039*** (−6.951)				
Ln(initial city size)			−0.064 (−1.616)	−0.099*** (−8.819)		
D(belong to an urban agglomeration)					0.791*** (12.107)	0.389*** (6.364)
D(proximity high-tier cities)	0.538*** (6.531)		0.744*** (11.007)		0.218* (1.855)	
Ln(population density)× D(proximity high-tier cities)		−0.006 (−0.987)				
Ln(initial city size)× D(proximity high-tier cities)				0.103*** (13.451)		
D(belong to an urban agglomeration)× D(proximity high-tier cities)						0.135*** (2.756)
Constant	6.439*** (37.080)	7.017*** (51.805)	6.181*** (20.163)	6.418*** (53.187)	5.449*** (65.051)	5.933*** (36.812)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.968	0.968	0.968	0.968	0.967	0.967
N	4560	4560	4544	4544	4560	4560

Note: t statistics in parentheses; ***, **, and * indicate significance at the 1 %, 5 %, and 10 % significance levels, respectively. We replace the proximity to high-tier cities with the top 10 % of cities, and the coefficients of proximity to high-tier in Model 1–3 are 0.538 (1 % significant), 0.201 (5 % significant), and 0.171 (non-significant), which are basically consistent with the results in Table 6. It still shows that the high degree of agglomeration in neighboring cities can promote a city's economic growth, indicating the robustness of our results.

This may have been caused by local talents, capital, and other factors being absorbed by neighboring regions. The explained variable in Models 4–6 was GDP per capita. The coefficients of ρ and λ and the indirect coefficient of population density were significant and positive. This was consistent with the aforementioned finding that intracity population agglomeration had positive spatial spillover effects, and economic growth in a city is higher when its neighboring cities have larger economies.

Based on Eqs. (6) and (7), we analyzed the impact of intercity population agglomeration on urban economic growth (Table 6). The results indicated that the high degree of agglomeration in neighboring cities positively affected a city's economic growth, suggesting that the borrowed size effect between cities was larger than the agglomeration shadow effect. In particular, cities with larger initial city sizes were more likely to borrow the scale effect of neighboring cities. It was important to note that high levels of agglomeration in neighboring cities had a weak negative effect on the agglomeration–growth within a city, which indicated there might exist an agglomeration shadow.

Models 1, 3, and 5 included the three agglomeration indicators, and the dummy variable of proximity to high-tier cities was included to reflect intercity agglomeration effects. The coefficient of the dummy variable of proximity to high-tier cities in the three models was positive (significant at the 5 % level in Models 1 and 3), which indicated that the high degree of agglomeration in neighboring cities can promote a city's economic growth. This finding was consistent with the study of Wang et al. (2022), who believed that peripheries adjacent to highly agglomerated cores may benefit from the transfer of urban residents. However, the impact coefficient of intracity agglomeration reduced by about 0.28 compared with the non-inclusion of the neighboring agglomeration element, similar to the results of Table 5. This finding was corroborated by the results of Models 7, 9 and 11.

Models 2, 4, and 6 further included the interaction term neighboring city agglomeration with intracity agglomeration. The results found that

the coefficient of $\text{Ln}(\text{initial city size}) \times \text{D}(\text{proximity})$ on per capita urban land revenue was 0.037 (significant at the 1 % level), while the $\text{Ln}(\text{population density}) \times \text{D}(\text{proximity})$ coefficient was not significant, and the coefficient of $\text{D}(\text{belong to an urban agglomeration}) \times \text{D}(\text{proximity})$ was even significantly negative. This suggested that cities with larger initial city sizes were more likely to borrow the scale effect of neighboring cities. The results of Models 8, 10, and 12 also validated the aforementioned findings.

We further investigated spatial variations in the effect of intercity population agglomeration on the economic growth of Chinese cities (Table 7). The results of our analysis revealed location and policy variations in this effect. That is, the positive effect of neighboring city population agglomeration was larger in the eastern parts of China than in the central and western parts. Additionally, the provincial capitals and planned cities are larger than other cities, and the effects of the policies implemented at the beginning of the reform and opening-up period have weakened.

Model 1 reveals regional variations in the influence of intercity population agglomeration on economic growth. When population density was controlled for, the correlation coefficient of the dummy variable for proximity to high-tier cities in the east was 0.538 (significant at the 1 % level), whereas in the center and west, the coefficient was significantly negative. Models 2–4 reveal similar variations in proximity to high-tier cities for provincial capitals versus nonprovincial capitals, coastal versus inland areas, and temperate versus other climatic zones. The coefficients for the effects of intercity population agglomeration were 0.538 and 0.539 for provincial capitals and temperate zones, respectively, whereas the coefficient for coastal areas was negative. Models 5–8 were used to examine the influence of intercity population agglomeration on urban economic growth, with initial city size controlled for. The influence coefficients for intercity population agglomeration in eastern regions and provincial capitals were 0.201 and 0.332, respectively, and the correlation coefficients for coastal and

Table 7
Spatial variations in the effect of intercity agglomeration on economic growth.

Core explanatory variables: Ln(population density) and D(proximity high-tier cities)	Location	Policy		Natural factor	Core explanatory variables: Ln(initial city size) and D(proximity high-tier cities)	Location	Policy		Natural factor
	Model 1 (xi:reg)	Model 2 (xi:reg)	Model 3 (xi:reg)	Model 4 (xi:reg)		Model 5 (xi:reg)	Model 6 (xi:reg)	Model 7 (xi:reg)	Model 8 (xi:reg)
Ln(population density)	0.592*** (70.301)	0.592*** (70.301)	0.592*** (70.301)	0.592*** (70.301)	Ln(initial city size)	0.134** (2.310)	−0.021 (−1.263)	0.134** (2.310)	0.112*** (3.369)
D(proximity high-tier cities)	−0.480*** (−8.840)	−0.535*** (−4.453)	0.538*** (6.531)	−0.781*** (−12.297)	D(proximity high-tier cities)	−0.146** (−2.426)	−0.063 (−1.165)	0.201** (2.048)	−0.939*** (−13.414)
D(proximity high-tier cities) × D(East)	1.018*** (19.658)				D(proximity high-tier cities) × D(East)	0.347** (2.576)			
D(proximity high-tier cities) × D(Middle)	−0.003 (−0.043)				D(proximity high-tier cities) × D(Middle)	−0.081 (−0.603)			
D(proximity high-tier cities) × D(provincial capitals and planned cities)		1.073*** (18.264)			D(proximity high-tier cities) × D(provincial capitals and planned cities)		0.332*** (3.212)		
D(proximity high-tier cities) × D(open coastal cities)			−0.925*** (−22.221)		D(proximity high-tier cities) × D(open coastal cities)			−0.404*** (−6.693)	
D(proximity high-tier cities) × D(temperate zone)				1.320*** (27.271)	D(proximity high-tier cities) × D(temperate zone)				0.621*** (6.896)
Control variables	Yes	Yes	Yes	Yes	Control variables	Yes	Yes	Yes	Yes
R ²	0.987	0.987	0.987	0.987	R ²	0.971	0.971	0.971	0.971
N	4560	4560	4560	4560	N	4544	4544	4544	4544

Note: *t* statistics in parentheses; ***, **, and * indicate significance at the 1 %, 5 %, and 10 % significance levels, respectively. The control variables were per capita urban land capital stock, number of patent applications per 10,000 persons, share of manufacturing employment, and average annual temperature. We also used GDP per capita as an explained variable to verify the spatial differences in intercity population agglomeration factors. The influence coefficient of the eastern city, provincial capital, and temperate zone proximity to high-tier cities stabilized at approximately 0.641 (significant at the 1 % level), and the coefficient of open coastal cities remained negative.

temperate zones were negative. Additionally, for eastern and provincial capitals and planned cities, analysis of the effect of spatial variations in intercity population agglomeration with GDP per capita as the explained variable indicated that such variations still exist.

Taken together, intracity population agglomeration had significant spatial spillover effects. The high degree of agglomeration in neighboring cities positively affected a city's economic growth and had a weak negative effect on intracity agglomeration–growth effects. This finding confirms that the borrowed size effect was larger than the agglomeration shadow effect among Chinese cities. Moreover, the promotion of intracity agglomeration was stronger than intercity agglomeration. Furthermore, spatial and policy differences affect the influence of intercity population agglomeration on economic growth.

4.3. Robustness tests

To test the robustness of our models, we first adopted the IV estimation approach to address potential endogeneity between population agglomeration and economic growth and the circular reasoning problem between density and per capita urban land revenue. We used the RDLS

and river network density as IVs for population density and slope as an IV for initial city size because these three physical geographic elements are objective exogenous variables that are unaffected by economic growth and time but influence urban population density and size levels (Peng et al., 2023). We used the interaction term of the physical geography factors and the inverse of the exchange rate of the external shock variable to construct panel data. Since the initial city size is time-invariant and could not be tested with panel instrumental variables, we replaced it with the permanent population from 2005 to 2020 in Table 8.

Table 8 presents the IV regression results of the influence of population agglomeration on urban economic growth. The Kleibergen–Paap rk LM values for all models were significant at the 1 % level, and the Kleibergen–Paap rk Wald F-test statistic exceeded the 10 % critical value of 16.38, indicating effective IVs. These results suggest that the correlation coefficients of the effects of population density on per capita urban land revenue and GDP per capita were stable at approximately 0.78 and 0.52, respectively, and the coefficients of intercity population agglomeration on economic growth were stable at approximately 0.04. The coefficients of initial city size on per capita urban land revenue and GDP per capita were still significantly positive. These results corroborate

Table 8
Results of instrumental variables tests.

Intracity agglomeration	Ln(RDLS×exchange rate)		River network density×exchange rate		Slope×exchange rate	
	Per capita urban land revenue Model 1 (2SLS)	Per capita GDP Model 2 (2SLS)	Per capita urban land revenue Model 3 (2SLS)	Per capita GDP Model 4 (2SLS)	Per capita urban land revenue Model 5 (2SLS)	Per capita GDP Model 6 (2SLS)
Ln(population density)	0.806*** (102.158)	0.230*** (8.385)	0.731*** (40.810)	0.666*** (5.380)		
Ln(initial city size)					51.449*** (3.333)	3.426*** (5.632)
Constant	1.795*** (15.212)	5.496*** (12.286)	2.643*** (11.639)	−1.469 (−0.593)	−293.714*** (−2.931)	−11.774*** (−2.877)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
City fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen–Paap rk LM statistic	729.09***	237.39***	243.78***	50.62***	10.19**	47.84***
Kleibergen–Paap Wald rk F statistic	1505.27	264.85	75.63	37.59	10.66	47.62
Stock–Yogo weak ID test critical values	16.38	16.38	16.38	16.38	16.38	16.38
N	4560	4560	4560	4560	4544	4544

Intercity agglomeration	Ln(RDLS×exchange rate)		River network density×exchange rate		Slope×exchange rate	
	Per capita urban land revenue Model 7 (2SLS)	Per capita GDP Model 8 (2SLS)	Per capita urban land revenue Model 9 (2SLS)	Per capita GDP Model 10 (2SLS)	Per capita urban land revenue Model 11 (2SLS)	Per capita GDP Model 12 (2SLS)
Ln(population density)	0.806*** (101.939)	0.230*** (8.382)	0.737*** (43.880)	0.638*** (5.443)		
Ln(initial city size)					51.640*** (3.331)	3.371*** (5.671)
D(proximity high-tier cities)	0.030*** (2.851)	0.026** (2.053)	0.043*** (3.582)	0.041* (1.778)	0.205 (0.727)	0.049** (2.143)
Constant	1.781*** (15.131)	5.481*** (12.238)	2.552*** (11.740)	−1.037 (−0.446)	−294.977*** (−2.932)	−11.466*** (−2.866)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
City fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen–Paap rk LM statistic	724.11***	237.6***	253.58***	51.85***	10.18**	48.738***
Kleibergen–Paap Wald rk F statistic	1495.48	264.88	88.75	40.02	10.64	48.556
Stock–Yogo weak ID test critical values	16.38	16.38	16.38	16.38	16.38	16.38
N	4560	4560	4560	4560	4544	4544

Note: z statistics in parentheses; ***, **, and * indicate significance at the 1 %, 5 %, and 10 % significance levels, respectively. Since the initial city size was time-invariant and could not be tested with panel instrumental variables, we replaced it with the permanent population from 2005 to 2020 in the instrumental variables test. Because the permanent resident population of Chongqing city was outlier, the data of Chongqing city were excluded from the model with initial city size, for which the sample size was 4544.

the robustness of our models and regression results.

Second, we assessed the municipal district-level effects, examining the possible mismatch between GDP, permanent population, and built-up area in city-wide statistics. Because the available Chinese statistical yearbooks did not provide data on the permanent population for municipal districts at the prefectural level for consecutive years, we constructed a panel of municipal districts only for 2000, 2010, and 2020.

Table 9 presents the regression results obtained using the municipal district data. The results reveal that the correlation coefficients for the effects of population density and initial city size on per capita urban land revenue were stable at approximately 0.85 and 0.77, and the coefficient of D (proximity to high-tier cities) was stable at approximately 0.5. These results indicate that the promotion of intracity population agglomeration was stronger than that of intercity population agglomeration. However, the results indicate that borrowed size and the agglomeration shadow coexist and that borrowed size exerts a greater effect than the agglomeration shadow does. Specifically, the correlation coefficients for the effects of population density on per capita urban land revenue and GDP per capita declined by approximately 0.2 and 0.1, respectively, with the inclusion of intercity agglomeration elements in the model. Additionally, the correlation coefficients of the effects of initial city size on per capita urban land revenue and GDP per capita were reduced by approximately 0.08 and 0.06, respectively.

Third, we adopted GDP modified for nighttime luminance data to address inaccuracies in GDP data from the China City Statistical Yearbook (Table 10). The results corroborated our findings: intracity population agglomeration had a substantial positive influence on urban economic growth, and the agglomeration shadow effect was present, but exceeded by the borrowed size effect, especially in cities with high initial size.

Models 1–4 reveal that the influence of population density and initial city size on modified per capita urban land revenue remained stable at approximately 0.87 and 0.03. Models 5–8 demonstrate that the influence of population density and initial city size on modified GDP per capita was significant. These results verified the strong influence of intracity population agglomeration on the economic growth of Chinese cities. Furthermore, the agglomeration of neighboring cities generally positively affected a city's economic growth. Models 9–12 reveal the influence of neighboring cities' agglomeration on a city's modified per capita urban land revenue, with a positive interaction effect of Ln(initial

city size) $\times$ D (proximity). The positive effect of neighboring cities and the interaction term coefficient in Models 13–16 were consistent with these results. These findings suggest that the borrowed size effect between cities is larger than the agglomeration shadow effect is, particularly for cities with large initial sizes. Furthermore, because the correlation coefficients of the effects of population density on modified per capita urban land revenue and GDP per capita decreased by 0.285 and 0.033, respectively, the results of Models 9–16 verified the influence of the agglomeration shadow effect on Chinese cities.

Finally, we used data on 35 provincial capitals and planned cities and 285 other cities, excluding the year 2020, to verify possible effects unique to large cities and economic instability due to COVID-19 (Table 11). The results demonstrate that our models and empirical results were robust. Models 1–6 present the regression results obtained using data from 35 provincial capitals and planned cities, suggesting that the correlation coefficients of three core variables—population density, initial city size, and the dummy variable for an urban agglomeration—were all significantly positive. Additionally, the effect of population density on GDP per capita remained significantly positive in the 35 provincial capitals and planned cities, suggesting an inconsistency between population density and GDP per capita in resource-based cities in China. Furthermore, Models 7–12 present the regression of population agglomeration on per capita urban land revenue for 285 cities from 2005 to 2019, revealing that intracity population agglomeration had a significant positive effect on urban economic growth, with the coefficient of intercity population agglomeration stable at approximately 0.24. Moreover, consistent with the results of Models 7–12, we measured the effects of population agglomeration variables on GDP per capita, revealing that population agglomeration had a positive effect on urban economic growth, whereas the influence coefficient of population agglomeration of neighboring cities on urban economic growth stabilized at approximately 0.48. Notably, when intercity agglomeration elements were added to the models, the influence coefficient of population density on per capita land revenue and per capita GDP were reduced by approximately 0.276 and 0.046, respectively.

After addressing the problems of endogeneity, possible inconsistencies in statistical indicators, and inaccuracies in the GDP reported in Chinese statistical yearbooks, we observed that intracity population agglomeration and intercity population agglomeration had a significant positive effect on economic growth, with the effect of

Table 9
Results on effect of agglomeration economy on economic growth in municipal district.

	Explained variable: per capita urban land revenue				Explained variable: GDP per capita			
	Intracity agglomeration		Intercity agglomeration		Intracity agglomeration		Intercity agglomeration	
	Model 1 (FE)	Model 2 (xi:reg)	Model 3 (xi:reg)	Model 4 (xi:reg)	Model 5 (FE)	Model 6 (xi:reg)	Model 7 (xi:reg)	Model 8 (xi:reg)
Ln(population density)	0.845*** (52.989)		0.613*** (23.867)		−0.073*** (−3.161)		−0.176*** (−7.184)	
Ln(initial city size)		0.768*** (8.957)		0.680*** (6.668)		0.359*** (6.346)		0.296*** (4.443)
D(proximity high-tier cities)			0.592* (1.866)	1.075 (1.485)			0.592* (1.866)	0.910** (2.065)
Ln(population density) $\times$ D(proximity high-tier cities)			0.069*** (3.216)				0.069*** (3.216)	
Ln(initial city size) $\times$ D(proximity high-tier cities)				−0.102 (−0.785)				−0.093 (−1.167)
Constant	0.709** (2.395)	−1.274*** (−3.144)	0.902** (2.093)	−1.014** (−2.432)	9.171*** (23.451)	4.335*** (12.096)	8.168*** (14.620)	4.520*** (12.561)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Hausman test p-value	0.0000				0.0000			
R ²	0.928	0.953	0.979	0.953	0.214	0.884	0.893	0.884
N	854	851	854	851	854	851	854	851

Note: *t* statistics in parentheses; ***, **, and * indicate significance at the 1 %, 5 %, and 10 % significance levels, respectively. The control variables were per capita urban land capital stock, capital stock per capita, number of patent applications per 10,000 persons, share of manufacturing employment, and average annual temperature. Because the permanent resident population of Chongqing city was outlier, the data of Chongqing city were excluded from the model with initial city size, for which the sample size was 851.

Table 10

Results of modified GDP data for nighttime lighting data.

Intracity agglomeration	Explained variable: per capita urban land revenue				Explained variable: GDP per capita			
	Population density		Initial city size		Population density		Initial city size	
	Model 1 (FE)	Model 2 (FE)	Model 3 (xi:reg)	Model 4 (xi:reg)	Model 5 (FE)	Model 6 (FE)	Model 7 (xi:reg)	Model 8 (xi:reg)
Ln(population density)	0.898*** (252.728)	0.878*** (243.133)			0.017*** (3.130)	−0.002 (−0.313)		
Ln(initial city size)			0.033*** (2.648)	0.028** (1.996)			0.036*** (4.356)	0.061*** (6.446)
Ln(per capita urban land capital stocks)	0.119*** (27.723)	0.120*** (28.912)	0.793*** (108.226)	0.793*** (108.372)				
Ln(capital stock per capita)					0.119*** (27.723)	0.120*** (28.912)	0.359*** (47.170)	0.356*** (46.842)
Patent applications per 10,000 persons	−0.002*** (−13.154)	−0.002*** (−8.980)	0.006*** (25.591)	0.006*** (25.613)	−0.002*** (−13.154)	−0.002*** (−8.980)	0.001*** (7.826)	0.002*** (8.590)
Share of manufacturing employment		0.362*** (10.671)		0.118** (2.523)		0.362*** (10.671)		0.220*** (7.098)
Average annual temperature		−0.056*** (−13.467)		0.025*** (3.310)		−0.056*** (−13.467)		−0.005 (−0.938)
Constant	0.010 (0.202)	0.877*** (10.662)	1.490*** (15.324)	1.231*** (10.655)	8.127*** (94.398)	8.984*** (85.509)	6.358*** (68.454)	6.226*** (60.784)
Hausman test p-value	0.0000	0.0000			0.0000	0.0000		
R ²	0.951	0.954	0.970	0.970	0.263	0.312	0.968	0.969
N	4560	4560	4544	4544	4560	4560	4544	4544

Intercity agglomeration	Explained variable: per capita urban land revenue				Explained variable: GDP per capita			
	Population density		Initial city size		Population density		Initial city size	
	Model 9 (xi:reg)	Model 10 (xi:reg)	Model 11 (xi:reg)	Model 12 (xi:reg)	Model 13 (xi:reg)	Model 14 (xi:reg)	Model 15 (xi:reg)	Model 16 (xi:reg)
Ln(population density)	0.613*** (74.863)	0.615*** (74.417)			−0.050*** (−9.582)	−0.048*** (−8.773)		
Ln(initial city size)			0.219*** (3.753)	−0.017 (−1.054)			0.014 (0.369)	−0.098*** (−9.009)
D(proximity high-tier cities)	0.722*** (9.007)		0.099 (1.003)		0.722*** (9.007)		0.657*** (9.953)	
Ln(population density) × D(proximity high-tier cities)		−0.009 (−1.518)				−0.009 (−1.518)		
Ln(initial city size) × D(proximity high-tier cities)				0.028** (2.459)				0.096*** (12.819)
Constant	0.487*** (3.528)	1.273*** (13.989)	−0.267 (−0.618)	1.363*** (9.736)	6.596*** (39.082)	7.376*** (56.042)	5.910*** (19.755)	6.687*** (56.791)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.987	0.987	0.970	0.970	0.969	0.969	0.969	0.969
N	4560	4560	4544	4544	4560	4560	4544	4544

Note: *t* statistics in parentheses; ***, **, and * indicate significance at the 1 %, 5 %, and 10 % significance levels, respectively. The control variables were per capita urban land capital stock, capital stock per capita, number of patent applications per 10,000 persons, share of manufacturing employment, and average annual temperature. Because the permanent resident population of Chongqing city was outlier, the data of Chongqing city were excluded from the model with initial city size, for which the sample size was 4544.

intracity population agglomeration exceeding that of intercity population agglomeration. We also demonstrated that the effect of borrowed size coexists with the agglomeration shadow effect and that the effect of borrowed size exceeded that of the agglomeration shadow effect, especially for cities with large initial sizes.

5. Discussion and conclusion

5.1. Discussion

We first verified that the intracity population agglomeration had a significant and robust positive impact on urban economic growth in China, in line with our [Hypothesis 1](#). This finding is consistent with findings from Europe and the United States that factors such as population density and city size promote economic growth ([Ciccone, 2002](#); [Duranton, 2016](#); [Faberman & Matthew, 2016](#); [Ottaviano & Pinelli, 2006](#)). This confirms that our basic regression results are reasonable. For example, [Ciccone \(2002\)](#) found that a doubling of urban employment density raises productivity in the United States and Europe by about 6 %

and 4.5 %, respectively. [Combes et al. \(2008\)](#) suggested that a doubling of employment density in France would raise wages by about 3.5 %. In addition, we found that this agglomeration–growth effect was moderated by spatial and policy-related factors. Specifically, this effect was stronger in the eastern and central regions of China and provincial capitals and planned cities. These findings are supported by those of other studies from China, India, and the United States. For example, [Chen, Li, et al. \(2018\)](#) reported that the relationship between population agglomeration and urban economic growth was the strongest to weakest in the eastern, central, and western regions of China, respectively. [Abhishek et al. \(2017\)](#) discovered that agglomeration most strongly promoted economic growth in Indian provincial capitals.

We also added a quadratic term for population density to verify whether there existed an inverted U-shaped relationship between population agglomeration and urban economic growth, and the results showed that there was still a significant positive linear relationship between them. Specifically, we estimated that the population density at the summit of the inverted U-shaped curve was 87,960 persons/km², while the highest population density in Chinese cities was less than

Table 11

Results of robustness tests for adjusting the object of study.

Object of study: 35 provincial capitals and planned cities in 2005–2020	Explained variable: per capita urban land revenue			Explained variable: GDP per capita		
	Model 1 (FE)	Model 2 (xi:reg)	Model 3 (xi:reg)	Model 4 (FE)	Model 5 (xi:reg)	Model 6 (xi:reg)
Ln(population density)	0.920*** (48.288)			0.051** (2.453)		
Ln(initial city size)		0.377*** (14.662)			0.288*** (4.099)	
D(belong to an urban agglomeration)			0.613*** (4.527)			0.683*** (8.309)
Constant	1.204*** (3.742)	0.612** (2.197)	2.674*** (9.718)	9.305*** (25.861)	5.310*** (8.195)	6.675*** (30.735)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Hausman test p-value	0.0000			0.0000		
R ²	0.919	0.971	0.971	0.121	0.960	0.962
N	560	544	560	560	544	560

Object of study: 285 prefecture-level cities in 2005–2019	Intracity agglomeration			Intercity agglomeration		
	Model 7 (FE)	Model 8 (xi:reg)	Model 9 (xi:reg)	Model 10 (xi:reg)	Model 11 (xi:reg)	Model 12 (xi:reg)
Ln(population density)	0.864*** (215.676)			0.588*** (69.110)		
Ln(initial city size)		0.034** (2.406)			0.102* (1.786)	
D(belong to an urban agglomeration)			0.557*** (4.672)			0.435*** (4.638)
D(proximity high-tier cities)				0.511*** (6.352)	0.215** (2.207)	0.185 (1.094)
Constant	1.202*** (13.818)	1.165*** (10.031)	0.861*** (4.157)	0.684*** (4.902)	0.449 (1.058)	0.797*** (10.616)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Hausman test p-value	0.0000					
R ²	0.943	0.971	0.971	0.987	0.971	0.971
N	4275	4260	4275	4275	4260	4275

Note: *t* statistics in parentheses; ***, **, and * indicate significance at the 1 %, 5 %, and 10 % significance levels, respectively. Because the permanent resident population of Chongqing city was outlier, the data of Chongqing city were excluded from the model with initial city size, for which the sample size was 544 and 4260, respectively.

80,000 persons/km². Relevant empirical evidence also found that the relationship between population density and urban economic growth was mostly linear (Chen, Li, et al., 2018; Liu, 2009). Scholars who found the inverted U-shaped relationship mostly used GDP per capita to measure urban economic growth, and the model had endogeneity problems.

Furthermore, we focus on the impact of intercity population agglomeration on urban economic growth in China and found that intracity population agglomeration had significant positive spatial spillover effects. The high degree of agglomeration of neighboring cities significantly promoted a city's economic growth. The borrowing size effect was stronger than the agglomeration shadow, which was in line with our Hypothesis 2. Our finding of a borrowed size effect is consistent with findings for second-tier cities in Western Europe (Camagni et al., 2015). For example, Ciccone (2002) found that the production density of neighboring regions had a significant positive effect on regional economic growth; Rice et al. (2006) reported that a doubling of the working-age population in the adjacent areas (within a 40-min drive) would increase the city's productivity by 3.5 % in the United Kingdom. Using microdata from 2007 to 2015, De Vos et al. (2020) demonstrated that small and medium-sized municipalities near large cities in Sweden achieved borrowed size through broadband network penetration, which enhanced labor market benefits.

Numerous empirical studies by Chinese scholars on the economic growth benefits of intercity agglomeration also support our conclusions. These studies suggest that small cities close to large cities are more positively influenced than negatively influenced by them. For example, Sun and Ding (2016) argued that there was a borrowed size effect in China's Yangtze River Delta region, with high-tier cities having a

positive spatial spillover effect on their neighboring cities; Li and Zhang (2021) also suggested that doubling the total population size of other cities within an urban agglomeration increase a city's income by 6.7 %–8.0 %. Sun et al. (2022) focused on small towns around large cities and discovered higher service function indices for towns near Shanghai. Additionally, they have also found that Chinese cities were in a mutually borrowed development pattern. The level of economics in neighboring cities had a strong positive spatial spillover effect on a city's economic growth, which means that there existed a spatial correlation of strong-strong association between cities. For example, Hong et al. (2010) documented interdependent economic growth in neighboring regions, with spatial spillovers from geographic proximity contributing to the spatial agglomeration of regions with similar economic levels. This finding is consistent with those of Yan et al. (2021), who observed significant positive spatial spillovers of urban economic growth in China. Another study also reported that small and medium-sized cities gained human and physical capital from larger cities and experienced knowledge spillovers (Yao & Wu, 2020). These studies add to the robustness of our conclusion that the borrowed size effect is stronger than the agglomeration shadow effect and demonstrate a spatial correlation of strong-strong association between cities.

We also discovered that intracity population agglomeration was more powerful than intercity population agglomeration as a driver of economic growth. This result is consistent with the study of Meijers et al. (2016), who have shown that the effect of local city size on the metropolitan functions was larger than the potential population size of the adjacent area in Europe (about 2.5 times). Similarly, Chen et al. (2023) demonstrated that the internal agglomeration effect was stronger than the external borrowed size effect was in the Guanzhong Plain urban

agglomeration in China, with the influence coefficient of internal agglomeration on economic growth of districts and counties stabilizing at approximately 0.12, whereas the coefficient of intercity population agglomeration was approximately 0.03. Moreover, [Lei and Jiao \(2023\)](#) revealed that intracity population agglomeration had a greater influence on promoting the economic growth of Chinese cities than intercity network externalities did, which are nevertheless becoming more prevalent.

It was worth noting that our study found that high levels of agglomeration in neighboring cities had a minor negative effect on the agglomeration–growth within a city. This is different from the view of [Ciccone \(2002\)](#). Using data from Europe, he documented that the coefficient of agglomeration effects within regions was unaffected by the inclusion of neighboring regions' agglomeration factors. This may be due to differences in regions, degree of agglomeration of neighboring cities, and city size. In fact, empirical studies have discovered a distinct of agglomeration shadow phenomenon in some regions of China. For instance, [Chen and Sun \(2017\)](#) found that the core mega-cities in the Beijing–Tianjin–Hebei urban agglomeration inhibit the economic growth of neighboring cities and that there is a poverty belt around Beijing and Tianjin. Additionally, [Wang and Tong \(2017\)](#) discovered that Beijing absorbed labor from small and medium-sized cities within a 200-km radius, reducing labor agglomeration in surrounding cities and indicating the existence of an agglomeration shadow in the Beijing metropolitan area. [Zhao et al. \(2022\)](#) explored the agglomeration shadow effect in the Pearl River Delta region and discovered pronounced features of agglomeration shadows in the remote towns of the Guangzhou–Foshan metropolitan area.

5.2. Conclusion

In this study, we constructed a theoretical framework and measurement model that describes the effects of both intercity agglomeration (specifically, borrowed size and agglomeration shadow effects) and intracity agglomeration on economic growth. Adopting a comprehensive set of indicators of agglomeration economy and economic growth and taking data from 285 prefecture-level cities in China, we verified the impact of population agglomeration on urban economic growth. The conclusions are as follows: first, intracity population agglomeration had a positive effect on urban economic growth. The agglomeration–growth effect was moderated by spatial and policy-related factors. Specifically, this effect was stronger in the eastern and central regions of China and provincial capitals and planned cities. Second, the borrowed size effect was stronger than the agglomeration shadow effect among Chinese cities. Intracity population agglomeration had significant spatial spillover effects. The high degree of agglomeration of neighboring cities positively affected a city's economic growth. Simultaneously, spatial and policy differences remain in this positive effect, with the borrowed size effect stronger in the east, provincial capitals and planned cities than in other cities. Third, the high levels of agglomeration in neighboring cities had a slight negative effect on the contribution of intracity agglomeration on growth. In particular, the impact coefficient of intracity agglomeration dropped by about 0.1, after incorporating elements of intercity agglomeration.

On the basis of these conclusions, this study makes three policy recommendations. First, continuously promote urban and rural planning based on spatial agglomeration. China has long been facing an allometric growth problem in which the speed of land urbanization was faster than the population urbanization, which caused a lot of unused built-up land and a low space utilization rate ([Li et al., 2009](#); [Yang et al., 2024](#)). Our findings indicated that intracity population agglomeration significantly promoted urban economic growth. Therefore, while promoting population urbanization, urban and rural planning should focus on land stock planning and constantly enhance the spatial agglomeration of population, industries, service industries. Second, construct a deeper intercity connectivity system. There has been a long-term economic

“competition and cooperation” between Chinese prefectural-level cities caused by regional barriers, which have hindered the flow of talent and capital between cities ([Liang et al., 2013](#)). Our findings found that a high degree of agglomeration of neighboring cities positively affected a city's economic growth. Thus, local governments need to build intercity connectivity mechanisms, such as building cross-city high-speed road systems and cross-city bridges. Third, large city functions should be spread out within urban agglomerations. The concentration of large city functions within the core cities of several Chinese urban agglomerations has led to economic homogenization and gradual widening of economic disparities between cities ([Li & Li, 2021](#)). Our findings demonstrate the feasibility of a development pattern under which large cities drive the growth of surrounding small cities through population agglomeration. Local governments should strengthen the mobility of talent, capital, and technology within urban agglomerations and rationally transfer industries from large cities to surrounding small and medium-sized cities to promote coordinated regional development.

This study has several limitations. First, this study focuses on the impact of population agglomeration on urban economic growth and does not comparatively study the impact of traditional industrial agglomeration on urban economic growth. Indeed, the process of urbanization complements the industrialization process, while industrial agglomeration and cross-industry spillover effects are key elements of modern urban development ([Pan et al., 2021](#)). Future studies can compare the differences in the impact of urban industry agglomeration and general urban agglomeration on urban economic growth. Second, our analysis did not account for the present-day phenomenon of relationships between cities becoming more akin to a network than a hierarchy ([Meijers et al., 2016](#); [Yang et al., 2022](#)), which may be a result of transportation or information links ([Yin et al., 2016](#); [Huang et al., 2020](#)). Therefore, future studies can use network statistics to analyze the effects of agglomeration. Last but not least, although we have measured the impact of urban agglomeration on inter-city economic growth with a series of datasets and indicator systems, we have not explored in depth the process of its impact and the explanatory mechanism, which is a vital part of our subsequent research.

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CRediT authorship contribution statement

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Declaration of competing interest

The article entitled “Population agglomeration, borrowed size and urban economic growth in China” has no conflict of interest in the submission of this manuscript.

Data availability

The data that has been used is confidential.

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